MIL-M-19633B(SHIPS)
10 March 1967
SUPERSEDING
MIL-M-19633A(SHIPS)
19 February 1964
(See 6.2)

MILITARY SPECIFICATION

MOTOR-GENERATOR, 60-CYCLE, AC TO 400-CYCLE AC

(VOLTAGE AND FREQUENCY REGULATED)

SHIPBOARD SERVICE

1. SCOPE

1.1 Scope.- This specification covers 60-cycle, alternating current (a.c.), to 400-cycle, a.c., voltage and frequency regulated motor-generator power supplies for Naval shipboard use.

1.2 Classification.- Motor generators and associated control equipment shall be of the following type and arrangements, as specified (see 3.5.2 and 6.1), for surface ships only.

Type I - General purpose.
Type II - Special purpose - with voltage balance regulators.
Arrangement A - With meters and switchgear.
Arrangement B - Without meters and switchgear.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearing, Ball, Annular, (General Purpose).
QQ-I-666	- Iron Castings, Malleable.
GGG-P-781	- Puller and Puller Kit, Mechanical and Mechanical Puller Attachment.

MILITARY

MIL-B-857	- Bolt, Nuts and Studs.
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917	- Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-D-1000	- Drawings, Engineering and Associated Lists.
MIL-D-1000/2	- Drawings, Engineering and Associated Lists.
MIL-I-1361	- Instruments, Electrical Measuring: Shunts, Resistors, and Transformers (Ship and Shore Use).
MIL-C-2212	- Controllers, Alternating-current, Naval Shipboard.
MIL-B-3743	- Brushes, Electrical, Contact; and Carbon Stock, Electrical Contact Brush.
MIL-C-5015	- Connectors, Electric, "AN" Type.
MIL-Q-9858	- Quality Program Requirements.
MIL-M-9868/1	- Microfilming of Engineering Documents, 35MM, For Naval Ship Systems.
MIL-M-15071	- Manuals, Equipment and Systems.
MIL-S-15083	- Steel Casting (Navy).
MIL-R-15109	- Resistors and Rheostats, Naval Shipboard.
MIL-S-16032	- Switch, Shipboard Alarm System.
MIL-M-16034	- Meter, Electrical-Indicating (Switchboard and Portable Types).
MIL-I-16104	- Indicators, Synchronization.
MIL-C-17361	- Circuit Breakers, Air, Electric, Insulated Enclosure (Shipboard Use).
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY (cont'd.)

MIL-C-17588 - Circuit Breakers (Automatic-ALB) and Switches, Toggle (Circuit Breakers, Non-automatic - NLB) Air, Insulated Inclosure, 125 Volts and Below AC or DC, Naval Shipboard.
MIL-S-18396 - Switches, Meter and Control, Naval Shipboard.
MIL-G-18709 - Grease, Ball and Roller Bearing.
MIL-W-19088 - Wattmeter, Switchboard Type, 4-1/2 Inch.
MIL-C-19836 - Cooler, Fluid, Industrial, Air Motor and Generator, Naval Shipboard.
MIL-H-22577 - Heating Elements, Electrical, Cartridge, Strip and Tubular Type.
MIL-V-23151 - Voltmeter, Expanded Scale Switchboard Type (Naval Shipboard Use).
MIL-M-23167 - Meter Frequency, Expanded Scale Switchboard Type (Naval Shipboard Use).
MIL-I-24137 - Iron Castings, Nodular Graphic (Ductile Iron) and Nodular Graphic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) (For Shipboard Application).

STANDARDS

MILITARY

MIL-STD-195 - Marking of Connections for Electric Assemblies.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

DRAWINGS

MILITARY

9000-S6202-73075 - Switch, Rotary, Multipole, Panel Mounted, 10 Amp, 120 VAC, Single Pole Blades, Type S-1JR.
9000-S6202-73295 - Switch, Rotary, Multipole, Type 4JR, Panel Mounted, 10 Amp, 120 VAC, Rotary Action Cutout.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd St., N.Y, N.Y. 10016.)

3. REQUIREMENTS

3.1 Definitions.-

3.1.1 Rated condition.- The characteristics assigned to a machine for a specified output rating such as, speed, voltage, frequency and current.

3.1.2 Nominal rating.- The normal constant operating condition or characteristic assigned to a machine when a range of operating conditions or characteristics are specified.

3.1.3 Continuous duty.- The requirement of service that requires satisfactory operation at any constant load in the range of no load to full load for an indefinitely long time.

3.1.4 Total harmonic content.- The total harmonic content of a complex wave is the total root mean square (r.m.s.) voltage remaining when the fundamental component is removed.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

3.1.5 Voltage modulation. - The cyclic variation or random dynamic variations, or both, about an average of the a. c. peak voltage during a steady state electric-system such as caused by voltage regulation and speed variations. The modulation envelope is formed by a continuous curve connecting each sine wave peak to the successive sine wave peak.

3.1.6 Synchronous impedance. - The per unit direct axis synchronous impedance equals the ratio of the field current at rated armature current on sustained symmetrical short circuit to the field current at normal open circuit voltage on the air-gap line.

3.1.7 Air-gap line. - The extended straight line part of the no-load saturation curve.

3.1.8 Deviation factor. - The ratio of the maximum difference between corresponding ordinates of the wave and of the equivalent sine wave to the maximum ordinate of the equivalent sine wave when the waves are superimposed in such a way as to make this maximum difference as small as possible.

3.1.9 Current modulation. - The cyclic variation or random dynamic variations, or both, about an average of the a. c. peak during a steady state electric-system such as caused by load and speed variations. The modulation envelope is formed by a continuous curve connecting each sine wave peak to the successive sine wave peak.

3.2 Performance requirements. -

3.2.1 The motor generator shall be designed to the following characteristics:

(a) Motor:
  (1) Voltage - 440 volts.
  (2) Phase - 3.
  (3) Speed - 1800 revolutions per minute (r.p.m.) synchronous.
  (4) Frequency - 60 cycles.
  (5) Type - Induction.
  (6) Voltage and frequency variation - The motor shall operate satisfactorly with an input voltage variation of plus or minus 10 percent and an input frequency variation of plus or minus 5 percent.

(b) Generator:
  (1) Kilowatt ratings (kw) - 5, 10, 15, 30, 60, 100, 200, or 300, as specified (see 6.1).
  (2) Output voltage - 450 volts.
  (3) Phase - 3.
  (4) Frequency - 400 cycles.
  (5) Power factor - 0.8 lagging.
  (6) Type - Rotating field, unless otherwise specified in the contract or order.

3.2.2 Duty. - Motor generators and control equipment shall be designed for satisfactory operation over the range of no load to full load when operating on the input voltage and frequency range specified in 3.2.1(a) (6).

3.2.3 Ambient temperature. - The motor generators and all associated control equipment shall be designed for operation in an ambient temperature of 50° Centigrade (C.) maximum.

3.2.4 Shock. - Motor generators and all associated control equipment shall be capable of withstanding the high impact shock test as specified for grade A class I equipment of MIL-S-901.

3.2.5 Insulation resistance. - The insulation resistance when connected in accordance with 4.6.5 shall not be less than 50 megohms for motor generators and 10 megohms for associated control equipment.

3.2.6 Grounding. - Equipment shall be designed to operate satisfactorily from an ungrounded ship's power system, but accidental grounding of this system shall not cause mal-operation. Design and construction shall be such that all exposed metal parts are at ground (ship's hull) potential at all times; however, none of the circuits of the equipment furnished shall be tied to ground. The output of the motor generato. power supply shall also be designed as an ungrounded system; however, the equipment shall be designed to perform in accordance with the requirements of this specification in the event of single failure to ground and when operating with loads which may have high impedance circuits to ground such as may be caused by electromagnetic interference suppression networks.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

3.2.7 Mode of operation. - The equipment shall be designed for automatic operation only.

3.2.8 Source of excitation power. - Generator excitation power may be obtained from rectified 400 cycle generator output. Power for flashing the field of the generator during startup may be obtained from the 60-cycle input power. Rotating exciters shall not be used unless the exciter output is rectified and fed directly to the generator field, eliminating the need for slip rings in the motor generator. Where remote sensing is specified (see 6.1), the sensing leads shall not carry the power to supply the regulator.

3.2.9 Voltage regulation. - Voltage regulation shall be achieved by a completely static regulator and excitation system. When tested in accordance with 4.6.16, voltage regulator shall be capable of maintaining the mean voltage within plus or minus 1/2 percent of rated voltage for any preset value within the range specified in 3.2.12 under the following conditions or any combination thereof:

(a) Load varying between no load and rated load.
(b) Load power factor varying between 0.8 lagging and unity.
(c) Variation of input voltage and frequency within the range specified.
(d) Variation of equipment temperature from cold at specified ambient to full load at specified ambient.
(e) Generator voltage adjusted to any point within the range specified in 3.2.12.
(f) Under the unbalanced condition specified in 3.2.20.1 (type I only).
(g) Under the unbalanced conditions specified in 3.2.20.2 (type II only).

3.2.10 Voltage recovery time. - The time to return and stay within plus or minus 0.5 percent of the rated value of the steady state voltage shall not exceed 0.1 second following change in an 0.8 p.f. load from no load to half load, from half load to full load, from full load to half load, and from three fourths load to one fourth load.

3.2.11 Voltage deviation (dip or rise). - Under the conditions specified in 3.2.10, the maximum dip or rise shall not exceed 9 percent of rated voltage.

3.2.12 Regulated voltage adjustment.- Provisions shall be made to adjust the regulated voltage within the range of 5.0 percent above and below rated voltage in steps of not more than 0.2 percent of rated voltage at any load from no load to 100 percent of rated load at rated power factor. The adjustment device shall have single knob control capable of changing output voltage of all three phases. This device shall be mounted inside of the enclosure for arrangement B in such a manner that it can be easily removed and relocated on a remote switchboard when required.

3.2.13 Frequency regulation. - Frequency regulation shall be achieved by the use of a completely static frequency regulator. When tested in accordance with 4.6.16, the frequency regulator shall maintain the output frequency to within plus or minus 0.5 percent of rated frequency for any preset value within the range specified in 3.2.16 under the following conditions or combination thereof.

(a) Load varying between no load and rated load.
(b) Load power factor varying between 0.8 lagging and unity.
(c) Variation of input voltage and frequency within the range specified.
(d) Variation of equipment temperature from cold at specified ambient to full load rise at specified ambient.
(e) Variation in general voltage as specified in 3.2.12.
(f) Generator frequency adjusted to any point within the range specified in 3.2.16.

3.2.14 Frequency recovery time. - The time to return and stay within plus or minus 0.5 percent of the rated value of the steady state frequency shall not exceed 0.5 second following sudden changes in an 0.8 p.f. load from no load to half load, from half load to full load and from full load to half load.

3.2.15 Frequency deviation. - Under the conditions specified in 3.2.14, the maximum frequency deviation shall not exceed 4 percent of the initial value of frequency.

3.2.16 Regulated frequency adjustment. - Provisions shall be made to adjust the regulated frequency within the range of plus or minus 1 percent of rated frequency in steps of not more than 0.1 percent at any load from no load to full load at rated power factor and with any voltage and frequency within the specified range applied to the motor. The adjustment device shall be mounted inside of the enclosure for arrangement B in such a manner that it can be easily removed and relocated on a remote switchboard when required.

3.2.17 Harmonic content. - From no load to full load, the r.m.s. value of the sum of all of the harmonic components in the output voltage wave shall not exceed 3 percent of the r.m.s. value of the rated

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

output voltage. The r.m.s. value of any single component shall not exceed 2 percent of the r.m.s. value of the rated output voltage. Harmonics of 0.1 percent or less, or the r.m.s. value of the rated output voltage may be disregarded in the determination of the harmonic content of the output voltage wave.

3.2.18 Amplitude modulation. - Amplitude modulation of the output voltage of the generator when operating with its intended voltage regulator shall not exceed one percent of the peak to peak voltage. This requirement shall apply for any loads between no load and rated load and at any power factor between 0.8 (lagging) and unity.

3.2.19 Deviation factor. - When tested in accordance with 4.6.27, the deviation factor of the 400-cycle voltage wave shall not exceed 5 percent.

3.2.20 Line unbalance. - The variation between line voltages for types I and II shall be as specified in 3.2.20.1 and 3.2.20.2.

3.2.20.1 Type I. - The variation between line voltages shall not be greater than 3 percent of rated voltage when a single phase load of 15 percent rated current at any power factor between 0.8 and unity is applied to any line of the generator which is already loaded with a balanced 3-phase load of 75 percent of rated current.

3.2.20.2 Type II. - The voltage variation between lines on 3-wire systems and between line and neutral of 4-wire systems shall not be greater than 1 percent under the unbalanced load conditions specified in table I.

Table I - Unbalanced loads.

Run	Line A	Line B	Line C	Load power factor
	Line current within ±5 percent	Line current within ±5 percent	Line current within ±5 percent	
1	0	0	0	0
2	Half	----	Full	0.8
3	Full	----	Half	.8
4	Full	Half	----	.8
5	Half	Full	----	.8
6	----	Half	Full	.8
7	----	Full	Half	.8
8	Full	Full	Full	.8

3.2.21 Voltage balance. - The difference between line voltages at the output terminals of the power supply (load side of circuit breaker) shall not be greater than 0.5 percent of rated voltage at no load or for any balanced 3-phase current up to rated current at any power factor between 0.8 and unity.

3.2.22 System short circuit performance. - The motor generator and control shall produce short circuit current to satisfy the following requirements

(a) $I^2t = 1100$

Where:
I = Sustained steady state generator output current (per unit value).
t = Time in seconds.

(b) The minimum value of generator short circuit current shall be 3 per unit.

(c) The peak value of short circuit current shall not exceed 11 per unit. The above requirement shall be met under the following 3-phase short circuit condition:

Initial load	Control	Generator field temperature
No load	Automatic	Hot

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

Hot field temperature shall be equivalent to normal full load rise obtained from the heat run plus the specified ambient temperature (50°C.). The design of the motor generator and control equipment shall be such that it can withstand a short circuit current of the magnitude required without damaging the equipment. Upon removal of the short circuit, rated voltage shall be automatically restored. No relays shall be used to provide excitation to obtain the specified short circuit characteristics.

3.2.23 Parallel operation. - Parallel operation shall be provided as follows:

(a) The equipment shall be designed so that any number (up to 4) of motor generators and control equipment of identical design will operate in parallel, as well as individually, in accordance with the requirements specified herein. Parallel operation shall be accomplished by cross-current compensation of the voltage and frequency regulators.
(b) The equipment shall be capable of being paralleled 180 degrees out of phase without damage.

3.2.23.1 Parallel operation performance. - The maximum difference in kilovar (kvar) and kw supplied by generators operating in parallel shall not exceed 10 percent of the kvar and 10 percent of the kw rating of one of the generators under any load condition between 20 and 100 percent of rated load. This requirement shall also apply for input voltage and frequency variation specified in 3.2.1(a) (6) variation in equipment temperature from ambient to full load rise, and variation in power factor between 0.8 lagging and unity. The design shall be such that the equipment will perform as specified herein when each power supply is connected to a different source and without any electrical interconnection between these sources:

(a) The maximum difference in kw supplied by the generators in parallel shall not exceed 5 percent of the kw rating of one of the generators.
(b) The maximum difference kvar supplied by the generators in parallel shall not exceed 6 percent of kvar rating of one of the generators.

3.2.24 Dielectric strength. - The motor generator and associated control equipment shall be capable of withstanding a dielectric strength test (see 4.6.6) of twice the normal voltage of the circuit plus 1,000 volts for one minute.

3.2.25 Remote sensing. - When remote sensing is specified (see 6.1), the following requirements shall be met at the remote sensing (load) point:

Voltage regulation (3.2.9 (a) thru (e)).
Frequency regulations (3.2.13).
Load rating (3.2.1 (b) (1)).
Regulated voltage adjustment (3.2.12).
Regulated frequency adjustment (3.2.16).
Short circuit performance (3.2.22).
Parallel operation (3.2.23).

3.2.25.1 The voltage drop between the generator terminals and the remote sensing point shall be between 0 and 7 percent of rated voltage based on the load. All other performance requirements shall be met at the output terminals of the equipment.

3.2.26 Electrical balance of motor. - The amperes in any phase of the motor at rated generator load shall not differ from the arithmetic average of the maximum and minimum phase amperes by more than 5 percent.

3.3 Detailed requirements. -

3.3.1 General requirement. - Equipment and components covered by this specification shall conform to MIL-E-917 unless otherwise specified herein. The equipment shall form a complete power supply consisting of a motor generator and its intended control equipment.

3.3.2 Reliability. - To meet the needs of the Naval service, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The supplier shall employ all reasonable methods possible in the process of manufacture to assure quality and maximum reliability consistent with the state of the art. In order to insure high equipment reliability under all service

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the current version before use.

conditions, avoid marginal applications, and provide adequate safety factors, it may be necessary to derate certain parts from their specification values when applying them to equipment circuits. The design shall include all possible features which will result in reliable and stable operation with minimum need for adjustment and alignment, minimum frequency of failure, and minimum requirements for maintenance. Maintenance shall be simplified so as to reduce the need for highly skilled maintenance personnel. The final product shall reflect the utmost in simplicity, have maximum reliability and be easy to install and maintain.

3.3.3 Interchangeability. - For this equipment, all essential parts including removable units, repair parts, and other items, shall be interchangeable physically and electrically with corresponding items or parts without selective matching, drilling, filing, or the use of undue force. Interchangeability of repair parts is necessary to permit both ship and shore maintenance to be performed by the Navy using repair parts stocked in the Naval supply system.

3.3.4 Life. - Wherever the expected life of parts is predictable and can be controlled through design or application, the parts (except bearings and brushes) shall be designed or applied so that equipment life of 40,000 hours of operation can be expected. The application or selection of bearings and brushes shall be such as to insure the following minimum hours of operation:

(a) Bearings - 5,000 hours (B-10) life.
(b) Brushes - 8,000 hours life.

3.3.5 Enclosures. - The motor generators (unless water cooled (see 6.1) and bulkhead mounted control equipment shall be furnished in dripproof protected 15 degree enclosures.

3.3.5.1 A dripproof protected 15 degree enclosure is an enclosure so designed and constructed that drops of liquid or solid particles falling on the enclosure at any angle not greater than 15 degrees from the vertical cannot interfere with satisfactory operation of the equipment within the enclosure. Louvers may be provided. All openings shall be constructed so as not to permit the passage of a rod 1/2 inch in diameter through the opening. Mounting bolt holes may penetrate the control equipment enclosure.

3.3.5.2 Water cooling. - When water cooling is specified (see 6.1), the motor generator and control cabinet thereon shall be furnished in a totally enclosed, water-air cooled enclosure. Bulkhead mounted equipment shall be furnished in dripproof protected 15 degree enclosures. A totally enclosed water-air cooled machine is one enclosed to prevent the free exchange of air between the inside and outside of the enclosure but not sufficient to be termed airtight. The machine is cooled by recirculating air through coolers conforming to class 2 of MIL-C-19836. The equipment shall operate satisfactorily with 85°F. water into the coolers. An air temperature alarm thermostatic switch conforming to type IC/N31A of MIL-S-16032 shall be provided and located to operate when the air leaving the cooler is 50°C. or more. Air filters shall be provided over the input to the coolers to remove carbon direct from the recirculating air. A drain plug shall be provided in the lowest part of the enclosure. An electric space heater shall be located in the lower part of the enclosure to maintain the temperature of the air within the enclosure 3 to 6 degrees centigrade above the outside ambient. Heating elements shall conform to type II (strip) or type III (tubular) of MIL-H-22577 and be suitable for operation on the motor generator input power. Adequate clearance between the heater and other parts shall be provided to avoid local overheating (temperatures above normal operating temperatures) of any parts. The heater shall be rigidly supported to prevent distortion under all operating conditions and the element sheath shall be grounded to the frame of the motor generator. The heater shall be interlocked with the motor starter so that the heater is automatically energized when the motor generator is stopped and deenergized when the motor generator is operating. Heater terminals shall be insulated or protected to prevent possible contact by personnel. Provision shall be included for deenergizing the heater circuit during disassembly and servicing. A warning plate shall be provided to warn personnel against servicing the motor generator without first deenergizing the heater.

3.3.6 Insulation class. - Insulation shall be class B or F.

3.3.6.1 If insulation classes are mixed within a motor generator, the thermal rating of the lesser temperature class of insulation shall not be exceeded.

3.3.7 Heating. - The motor generators and control equipment shall be so designed that the temperature rise do not exceed those specified in table II when tested in accordance with 4.6.22.2.

Source: https://assist.dla.mil -- Downloaded: 20260-19T06:25Z
Check the source to verify that this is the current version before use.

Table II - Maximum permissible temperature rise (degrees C.)
(based on 50° C. ambient - continuous rated load)

Name of part[1/]	Insulation Class	
	B	F
Bearings	60	60
Coils and coil connections - measured by thermocouple or thermometer	60	85
Measured by resistance	70	95
Cores and mechanical parts adjacent to insulation	60	85
Collector rings	75	100
Air - within 1/2 inch of meters	15	15
Capacitors - Air with 1/2 inch of case	35	35
Semi-conductor devices[2/]	25	25
Rheostats and resistors:		
Bare resistor material	375	375
Resistor embedding material	300	300
Bolted connections and terminal studs:		
Not plated	50	50
Silver plated	65	65
Contact:		
Solid	70	70
Solid, silver faced	80	80

1/ Where no measurement method is indicated, thermometer or thermocouple method is used.

2/ Silicon rectifiers, voltage reference elements, voltage regulator elements and transistors - permissible temperature rise on these parts is dependent upon operating conditions. Maximum permissible case temperature shall be shown on the equipment drawings (see 3.7).

3.3.8 Material restrictions. - The minimum requirements for materials to be used shall be as specified in table III.

Table III - Minimum material requirements

Item	Material	Specification	Limitations
Brushes	Various	MIL-B-3743	See MIL-B-3743
Caps, ball bearing end brackets, terminal boxes, covers, frames flanges, vee rings, shells, drip covers baffles, hand hole as access covers	Steel Malleable iron Nodular graphitic iron	See note 1 MIL-I-24137	QQ-I-666
Eye bolts	Steel		

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Table III - Minimum material requirements (cont'd.).

Item	Material	Specification	Limitations
Fans	Steel Aluminum	See note 1	
Lubrication	Grease	MIL-G-18709	Bearings operating at 110°C. and below
Holders, brush bodies, frames, fingers	Brass Bronze Bronze Steel		Corrosion resistant steel
Springs	Steel, carbon Steel, corrosion resistant Copper-beryllium		Cadmium plated and baked after forming
Shafts	Steel		Tensile strength not less than 75,000 p.s.i. and elongation not less than 20 percent in 2 inches
Spider and quills Studs, brush holder	Steel Steel	See note 1 See note 1	 Treated for corrosion resistance

Note 1.-Unless otherwise indicated in table III, steel parts may be cast, fabricated, wrought or forged. Cast steel shall be in accordance with grade B of MIL-S-15083, or equivalent, except that radiographic and magnetic particle inspection will not be required.

3.3.9 Lifting provisions. - Motor generators, control panels, and separately mounted control components weighing over 150 pounds shall be provided with eyebolts or other suitable means for hoisting and handling.

3.3.10 Identification and information plates. - The identification and information plates shall be attached to the part of the machinery or equipment which will not ordinarily be renewed during its normal service life. These plates shall be located where they can be read at all times without danger to personnel. Identification plates and information plates shall be made of brass, nickel-copper-alloy or corrosion resistant steel. The markings shall be either etched, stamped, engraved, metal-photo on anodized aluminum or cast in such a manner as to produce permanent and durable markings to last the anticipated life of the equipment. All etchings, engravings or stampings shall not be less than .003 inch deep. The characters on cast plates shall be raised to at least 0.03 inch. All engraved stamped or direct etched markings shall be filled with black paint enamel or lacquer (see 3.4.26 and 3.5.6 for details of information to be shown on the identification plates).

3.3.11 Inclined operation. - The motor generator and control equipment shall be designed to operate within specification limits under the following conditions:

(a) Motor generators:
  (1) Motor generator shaft inclined 15 degrees, generator end low for 0.5 hours.
  (2) Motor generator shaft inclined 15 degrees, motor end low for 0.5 hours.
(b) Control equipment inclined 15 degrees in any direction for 0.5 hours.

3.4 Motor generator. -

3.4.1 General requirement. - The motor generator shall be of the two bearing type with the stationary component parts of the motor and the generator rigidly coupled together in such a fashion that the combined unit will be capable of maintaining proper alignment without dependence upon the supporting structure of the

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

ship, and shall not require any additional structural bedplate. The rotating elements of the motor and generator shall be mounted on a single one piece shaft.

3.4.2 Mounting. - The motor generator shall be designed for horizontal deck mounting. The size of the mounting bolts shall be selected by the manufacturer in accordance with table IV and the following:

(a) Mounting bolts shall be grade 2 of MIL-B-857.
(b) Minimum bolt size shall be 3/4 inch.
(c) Maximum bolt size shall be 1-1/2 inches.
(d) Mounting bolt holes shall be round.
(e) Bolt hole diametrical clearance shall be 1/32 inch for 3/4 inch bolts and 1/16 inch for bolts over 3/4 inch.

3.4.3 Dimensions and weight. - The motor generators shall be as light as possible consistent with specified performance and design. The maximum dimensions and weight shall be as specified in table IV.

Table IV - Dimensions and weights

MG KW	Type	Control arrangement	Motor generator Max. inches and pounds				Control cabinet Max. inches and pounds				Voltage balancer Max. inches and pounds				MG mounting dimensions 3/ Max. inches	
			L	W	H 1/	Wt.	D 2/	W	H 1/	Wt.	D	W	H 1/	Wt	L	W
5	I	A	36	20	30	900	12	28	36	300					18	15
5	I	B	36	20	30	900	12	28	36	300					18	15
5	II	A	36	20	30	900	12	28	48	500					18	15
5	II	B	36	20	30	900	12	28	48	500					18	15
10	I	A	42	24	40	1800	12	28	36	300					21	18
10	I	B	42	24	40	1800	12	28	36	300					21	18
10	II	A	42	24	40	1800	12	28	48	500					21	18
10	II	B	42	24	40	1800	12	28	48	500					21	18
15	I	A	44	24	40	2000	12	28	36	300					22	18
15	I	B	44	24	40	2000	12	28	36	300					22	18
15	II	A	44	24	40	2000	12	28	48	600					22	18
15	II	B	44	24	40	2000	12	28	48	600					22	18
30	I	A	52	28	46	3000	12	28	36	300					26	21
30	I	B	52	28	46	3000	-	-	-	-					26	21
30	II	A	52	28	46	3000	12	28	36	300	16	28	48	500	26	21
30	II	B	52	28	46	3000	16	28	48	500					26	21
60	I	A	64	32	56	5000	12	28	36	300					32	24
60	I	B	64	32	56	5000	-	-	-	-					32	24
60	II	A	64	32	56	5000	12	28	36	300	16	28	60	600	32	24
100	I	A	68	36	60	7000	12	28	36	400					18/18	27
100	I	B	68	36	60	7000	-	-	-	-					18/18	27
100	II 4/	A	74	36	60	8000	12	28	36	400					20/20	27
100	II 4/	B	74	36	60	8000	-	-	-	-					20/20	27
200	I	A	84	46	74	12000	12	28	36	400					20/20/20	36
200	I	B	84	46	74	12000	-	-	-	-					20/20/20	36
200	II 4/	A	92	46	74	14000	12	28	36	400					25/25/25	36
200	II 4/	B	92	46	74	14000	-	-	-	-					25/25/25	36

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the currentversion before use.

Table IV - Dimensions and weights (cont'd.).

MG KW	Type	Control arrange-ment	Motor generator Max. inches and pounds				Control cabinet Max. inches and pounds				Voltage balancer Max. inches and pounds				MG mounting dimensions 3/ Max. inches	
			L	W	H 1/	Wt.	D 2/	W	H 1/	Wt	D	W	H 1/	Wt.	L	W
300	I	A	92	46	74	15000	16	28	48	500					22/22/22	36
300	I	B	92	46	74	15000	-	-	-	-					22/22/22	36
300	I 4/	A	100	46	74	17000	16	28	48	500					25/25/25	36
300	I 4/	B	100	46	74	17000	-	-	-	-					25/25/25	36

1/ Does not include removable lifting provisions.
2/ Does not include guard rail.
3/ 100 KW motor generator has 6 mounting bolt holes, length is between holes.
200 KW and 300 KW motor generators have 8 mounting bolt holes, length is between holes. Mounting shall be physically located about the center of the motor generator.
4/ Water cooled rotating unit, when specified (see 6.1).

3.4.4 Airborne noise. - Motor generators shall be designed to produce minimum airborne noise. The airborne noise limits, instrumentation and measurements shall be in accordance with MIL-STD-740 and the following:

Grade - Grade D equipment.
Operating condition - No load to full load at rated input voltage specified herein.

3.4.5 Internally excited vibration. -

3.4.5.1 Mechanical balance. - Motor generators shall be balanced at any load in the operating range. Windings not in mechanical symmetry shall have dummy coils. In general, the proper mechanical balance shall be effected by the use of balance weights securely attached, (if bolts are used, corrosion-resistant bolts and locknuts shall be required), removal of material, securely welded steel weights, or by metal carried in a receiver in such a manner as to preclude its breaking loose.

3.4.5.2 Balance limits. - The maximum allowable amplitude in inches shall be as specified in table V.

Table V - Mechanical balance

Weight of motor generators	Maximum allowable peak to peak amplitude
	Balance
Pounds	Inch
Up to 350	0.0005
From 351 to 800	.00075
From 801 to 2,000	.001
From 2,001 and up	.0015

3.4.6 Bearings. -

3.4.6.1 Radial, single row, both races extended, sealed, cartridge type, type 120 in accordance with FF-B-171 shall be used. A plate bearing the warning "DO NOT LUBRICATE" shall be secured to the end brackets.

3.4.6.2 Series. - Ball bearings shall be of medium series.

3.4.6.3 Mounting diameters and tolerances. - The shaft, shoulder, housing, and housing shoulder diameters shall be in accordance with the values as specified in table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Table VI - Bearing shaft, shaft shoulder, housing and housing shoulder diameters.

Bearing	Shaft diameter		Shaft shoulder diameter minimum	Housing diameter		Housing shoulder diameter maximum
	Minimum	Maximum		Minimum	Maximum	
305	0.9844	0.9847	1.140	2.4409	2.4416	2.280
306	1.1812	1.1815	1.340	2.8346	2.8358	2.620
307	1.3781	1.3785	1.590	3.1496	3.1503	2.940
308	1.5749	1.5753	1.813	3.5433	3.5442	3.310
309	1.7718	1.7722	2.000	3.9370	3.9379	3.660
310	1.9686	1.9690	2.270	4.3307	4.3316	4.000
311	2.1655	2.1660	2.468	4.7244	4.7253	4.380
312	2.3623	2.3628	2.688	5.1181	5.1191	4.750
313	2.5592	2.5597	2.968	5.5118	5.5131	5.190
314	2.7560	2.7565	3.156	5.9055	5.9068	5.560
315	2.9529	2.9534	3.380	6.2992	6.3005	5.940
316	3.1497	3.1502	3.550	6.6929	6.6942	6.250
317	3.3466	3.3472	3.840	7.0866	7.0879	6.620
318	3.5434	3.5440	4.060	7.4803	7.4817	7.000
319	3.7403	3.7409	4.290	7.8740	7.8754	7.380
320	3.9371	3.9377	4.488	8.4646	8.4660	7.940

3.4.6.4 Bearing pre-load washers or springs. - The use of bearing pre-load washers or springs or both are preferred provided the washer or spring, or both is secured in the housing to preclude its falling out of the end bracket during repairs and maintenance of the motor generator.

3.4.6.5 Mounting. - The bearings shall be mounted as follows:

(a) Fixed method - When the bearings are secured to the shaft by means of locknuts and lockwashers the following requirements shall apply:
  (1) One bearing shall be fixed axially in the housing by clamping, utilizing suitable housing and end cap shoulders.
(b) Opposed shoulder method - When the opposed shoulder method is used, the following information shall be submitted to the command or agency concerned for approval:
  (1) The initial end play required (to compensate for thermal expansion differentials between shaft and housing).
  (2) The method used to measure end play.

3.4.6.6 Housing construction. - The bearing housing shall be constructed to permit ready removal of the end bracket without the necessity of removing the bearings from the shaft. The housing shall have sufficient metal to permit redrilling and bushing.

3.4.6.7 Removal of bearing. - The design shall be such as to permit removal of the ball bearing by pulling on the inner ring of the bearing with one of the tools contained in a "complete" Naval shipboard set of tools specified in GGG-P-781.

3.4.7 Fans. - Fans shall be keyed to the shaft to prevent rotation and shall be secured in a manner to prevent axial displacement of the fan along the shaft. Shoulders, lockwashers and nut, sleeves or clamping bolts (in the case of split fan) may be used. Shrink rings or press rings may be used where the fans do not have to be removed from the shaft during disassembly of the motor generators. Locked set screws may be used only where closed end key seats or woodruff keys are used.

3.4.8 Cores. -

3.4.8.1 Mounting. - Cores shall be secured to the shaft, spider, or frame so as to prevent their axial and rotational displacement under operational or test conditions. A pin through the shaft is not an acceptable means of preventing axial displacement. Welding shall not be used to secure the core or quill to the shaft. The use of spiders constructed of individual flutes welded to the shaft is acceptable when preheat and stress relieving procedures satisfactory to the command or agency concerned are employed. Press or shrink fits without keys shall be approved by the command or agency concerned.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

3.4.8.2 Accessibility. - To remove or insert the rotor it shall be unnecessary to remove any stationary windings, remove any of the motor generator mounting bolts, or disassemble the rotor.

3.4.9 Collector rings. - Collector rings shall run true, have a smooth polished surface, and be free of porosity or hard spots. The rings with their insulation shall be shrunk over a steel sleeve and shall be of such thoroughly solid construction throughout as to insure against their becoming loosened or eccentric due to changes in temperature in service. The collector rings shall be secured rigidly to the shaft or spider in a manner as to positively prevent rotational or axial movement with respect to the shaft. Keys shall be used to prevent rotational movement. Axial movement shall be prevented by the use of locknuts or an interference fit. When an interference fit is used to prevent axial movement, the minimum interference shall be as specified in table VII. Where collector rings are located outboard of the bearings, axial movement shall be prevented by the use of a nut with a locking device. The wearing depth of collector rings shall be as shown in table VIII.

Table VII - Minimum interference fits for collector ring assembly shell to shaft.

Shaft diameter at assembly section	Minimum interference fits
Inches	Inch
Over 0 to 2, inclusive	0.0003
Over 2 to 3, inclusive	.0006
Over 3 to 4, inclusive	.0009
Over 4 to 5, inclusive	.0013
Over 5 to 6, inclusive	.0015
Over 6 to 7, inclusive	.0019
Over 7 to 8, inclusive	.0024
Over 8 to 9, inclusive	.0028
Over 9 to 10, inclusive	.0037
Over 10 to 11, inclusive	.0042
Over 11 to 12, inclusive	.0046
Over 12 to 13, inclusive	.0051
Over 13 to 14, inclusive	.0056
Over 14 to 15, inclusive	.0061
Over 15 to 16, inclusive	.0065
Over 16 to 17, inclusive	.0070

Table VIII - Wearing depth of collector rings.

Diameter (inches)	Permissible reduction in diameter
	Inch
2 and under	3/16
Over 2 and up to 4	3/8
Over 4 and up to 6	1/2
Over 6 and up to 8	5/8
Over 8 and up to 12	3/4
Over 12 and up to 15	7/8
Over 15	1

3.4.10 Brush rigging. -

3.4.10.1 Accessibility. - Brush holders shall be readily accessible for adjustment, replacement or renewal of brushes and springs. It shall not be necessary to remove the end brackets to replace or adjust the brush holders or brushes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

3.4.10.2 Mounting of collector ring brush holders. - Collector ring brush holders may be mounted on insulated studs attached to the end brackets.

3.4.10.3 Alignment. - Brush holder shall be so mounted and aligned that brushes are parallel to the shaft and do not extend beyond the edge of the collector ring.

3.4.10.4 Clearance. - The brush holder or brush rigging shall be adjustable so that the optimum clearance between the brush holder and collector ring may be maintained throughout the useful life of the collector ring.

3.4.11 Brush holders. -

3.4.11.1 General requirement. - Contact surfaces shall be of such material as to not react chemically to produce corrosion and sticking of brushes when brushes or graphite, electro-graphite or copper-graphite are used. The brush holders shall be of such design that the brushes will not bind in the brush holders. The brush-feeding device shall ensure proper brush contact under all normal conditions of brush and collector ring wear.

3.4.11.2 Brush stop. - Means shall be provided to prevent brushes from being thrown out of their brush holders as a result of severe shock or from being displaced to such a position that they may bind in their holders and not return to their normal operating position. The devices used shall not complicate replacement of the brushes, require the use of a specially designed brush, or result in cumbersome or intricate design of brush holders.

3.4.11.3 Pressure finger. - The pressure finger shall be of such design that when operating with a brush having a smooth top surface (not notched) or a hammer clip, it will not slide off the brush or clip under any operating conditions. When the brush is omitted from the holder, a stop shall prevent the finger from making contact with the collector ring.

3.4.11.4 Brush holder spring. - Springs shall be of either the adjustable type or constant pressure type. Where adjustable springs are used, means shall be provided for adjusting the brush pressure without special tools. Where constant pressure springs are used they shall be such that the brush pressure parallel to the length of the brush will be automatically maintained within 95 to 105 percent of average value throughout the useful life of the brush.

3.4.11.5 Types. - Brush holders shall be the full box type.

3.4.11.6 Terminal screw. - A separate terminal screw shall be provided for securing the brush shunt terminal lugs to the brush holder.

3.4.12 End brackets. - An accurately machined shoulder joint shall be provided between the frame and the end brackets. End brackets shall be secured to the frame by not less than four hexagon-head bolts. Resilient gaskets shall not be placed between any bearing support member and the frame. Ready access shall be provided to permit measurement of shaft speed.

3.4.13 Access holes. - Openings of adequate size and number shall be provided so as to give easy access to and a direct view of collector rings and brushes while the machine is in operation. When collector (slip) rings are used, it shall be possible to resurface them through access openings using commercially available, precision grinding and turning tools. These openings shall be provided with substantial transparent plastic covers and fastenings. The enclosing covers (hand-hole or access-hole covers) shall be readily removable. The engaging bolts or locking devices shall be attached to the frame, the end bracket, or the cover as the design permits, in such a manner as to prevent the bolts or locking devices from being misplaced or dropped into the machine during repair operations.

3.4.14 Feet. - Feet shall be machined on the mounting surfaces and shall be either cast integral with or welded to the frame and shall be provided with round holes for holding-down bolts.

3.4.15 Field poles and shoes. - Field poles and shoes may be an integral part of the complete rotor lamination and need not be individually removable. The method of securing rotating field poles to the shaft or spider shall result in a factor of safety of not less than 2.0, based on the elastic limit of the material and the rated speed of the motor generator.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

3.4.16 Field coils. - Field coils shall be thoroughly insulated from adjacent conducting and grounded parts. All formed coils shall be completely 1/2 lap wrapped with tape before the varnish dip and bake. All field coils in the same machine shall be readily interchangeable. The coils shall be so secured that they cannot become loosened or damaged by vibration, or produce vibration by a slight shifting in the coils position.

3.4.17 Coil connections. - All coil connections shall be made on the collector ring end of the motor or generator.

3.4.18 Excitation requirements. - The motor and generator design, construction, assembly, and quality control exercised shall be such that the ampere turn requirements for the motor and generator at any load do not differ more than plus or minus 10 percent from the design or nominal value.

3.4.19 Stray field reduction. - To reduce the stray magnetic field the motor generator shall incorporate the following:

(a) Stray magnetic fields caused by electrical circuits shall be kept to a minimum by making the area of electrical current loops as small as possible and arranging adjacent loops so that their fields will be in opposition. Current loops shall enclose no magnetic material unless essential for proper performance of equipment. All motor and generator windings and connections shall be balanced for ampere-turns around the shaft to eliminate stray fields insofar as possible. Each circuit, which makes even part of a turn around the machine shall return on itself back to the starting point, with conductors for the forward and reverse currents arranged at the same end of the machine and as close as adequate insulation permits. All connections shall be as short as possible.

3.4.20 Electromagnetic interference. - The equipment shall be designed to insure a minimum amount of electromagnetic interference.

3.4.21 Terminal boxes. - Terminal boxes shall be provided and shall be securely bolted or welded to the frame. The boxes shall not have any openings but shall be suitable for the drilling and installing of cable clamps or stuffing tubes by the installing activity. Terminal boxes not readily removable from the motor generator shall be provided with removable plates for the purpose of drilling as specified herein. Where the controls are mounted on top of the motor generator, separate terminal boxes shall not be furnished, but all connections shall be made in the control cabinet on the top of the motor generator. Removable cable entrance plates shall be provided on both sides of the control cabinet. Gaskets shall be furnished under all terminal box covers, removable plates, and between the box and motor generator frame, if the box is bolted to the frame.

3.4.22 Terminal boards. -

3.4.22.1 Generators having a rating of 10 kw or less. - A lead clamp shall be provided for the terminal leads. The terminal leads shall not be attached to the end brackets but shall be attached to the frame instead and shall extend to at least 6 inches beyond the point of support. The methods of fastening shall be such that no strain from the outside can come upon the connections within the generator frame. The use of unclamped rubber bushings is not a suitable method of securing such leads. The end of each generator lead shall be fitted with a connector of either the solder or the solderless type which shall be satisfactory to the command or agency concerned. Terminal lead holes in the generator frames shall have all edges rounded to prevent injury to lead insulation. The leads shall be enclosed in a terminal box which shall be solidly attached to the generator frame. A gasket shall be placed between the terminal box and the frame. The joints of the terminal box shall be of dripproof construction.

3.4.22.2 Generators having a rating greater than 10 kw. - All motor generator leads shall terminate at a terminal board within the terminal box. The terminal board shall be securely mounted within the terminal box in such a manner as to permit connection of the internal leads to the back of the board and the external leads to the front of the board. Terminal studs shall be secured to the boards in a manner to prevent turning of the studs when tightening or loosening the nuts used to secure lead connectors. Terminal identification corresponding to the lead identifications specified in MIL-STD-195 shall be permanently marked on the f. nt of the terminal board.

3.4.23 Terminal leads. - All terminal leads shall be of flexible, stranded cable or wire. An insulating sleeve or other suitable means shall be provided to prevent abrasion of the lead insulation by metallic edges of the motor generator frame, terminal box, piping or cable clamps.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

3.4.24 Securing terminal leads. - Leads shall be secured to prevent their coming in contact with moving parts or being chafed by contact with stationary parts. Leads shall not be secured to end brackets or in any other manner which would prohibit ready removal of the end brackets or rotating assembly. Terminal leads shall be secured to the terminals on the terminal board, brush holders, and collector rings by means of solderless connectors and jam nuts or lock washers or nuts to prevent the connectors from loosening under shock or vibration. Securing of leads to brush holders by means of special provisions in the design of the brush holder will be considered as an exception subject to approval of the command or agency concerned.

3.4.25 Lead identification. - All motor and generator leads shall be permanently marked in accordance with MIL-STD-195. Stamping of identification on solderless terminals is not permitted. In order to achieve standard marking of all leads, all motor generators shall rotate clockwise when facing the generator end. The direction of rotation shall be permanently marked on the frame of the motor generator by means of an instruction plate stating "rotation" with an arrow in the direction of rotation.

3.4.26 Identification plate markings. - A separate plate for the motor and the generator or a single larger plate may be furnished. The minimum data to be marked on the identification plate (or plates) shall include the following:

(a) Major unit name (that is, a.c./a.c. motor generator).
(b) Federal stock number.
(c) Manufacturer's name and serial number.
(d) Manual number.
(e) Contract number and year of manufacture.
(f) Blank space for Government inspector's official acceptance stamp.
(g) Salient design characteristics of motor, namely, horsepower, phase, voltage, current, speed, class of insulation and maximum ambient temperature.
(h) Salient design characteristics of generator, namely, kw., power factor, voltage, current, frequency, number of phases, class of insulation and maximum ambient temperature.

3.4.27 Permanent marking. -

3.4.27.1 Magnet frame. - The manufacturer's name, serial number and the Federal stock number (where available) shall be stamped underneath the identification plate in the solid metal of the magnet frame.

3.4.27.2 Rotating element. - The rotating element shall be marked with the manufacturer's serial number and Federal stock number (where available) shall be stamped on the shaft.

3.5 Control equipment. -

3.5.1 General requirements. - Control equipment requirements, in addition to those which affect the combined system performance and the detailed requirements of 3.3 shall be as specified herein.

3.5.2 Arrangements. - Control equipment shall be furnished in either one of the following arrangements, as specified (see 6.1):

(a) Arrangement A - A complete motor generator set in which metering devices and control switchgear is furnished with the motor generator and mounted in the bulkhead mounted control cabinet.
(b) Arrangement B - A motor generator set furnished without metering devices and control switchgear. The meters and switchgear for arrangement B motor generator sets will be furnished by a separate agency and located on a ship's main or auxiliary switchboard.

3.5.2.1 Control panel mounted equipment. - The following control equipment shall be furnished as specified and located in accordance with 3.5.2.2 and 3.5.2.3:

(a) Motor controller or starter (see 3.5.4.7).
(b) Voltage regulator and static exciter.
(c) Frequency regulator.
(d) Motor controller overload relay reset.
(e) Voltmeter for a.c. generator output[1/].
(f) Ammeter for a.c. generator output[1/].
(g) Frequency meter for a.c. generator output[1/].

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

3.5.3 Enclosures. -

3.5.3.1 Mounting. - The control equipment enclosures, unless mounted on the rotating unit, shall be designed for bulkhead mounting. The enclosures shall not be supported from more than one plane. The number, size and spacing of the mounting bolts shall be selected by the manufacturer in accordance with the following:

(a) Mounting bolts shall be grade 2 of MIL-B-857.
(b) Minimum bolt size shall be 5/8 inch.
(c) Maximum bolt size shall be 1.0 inch.
(d) Number of bolts shall be 4, 6, or 8; however, the minimum number required shall be used.
(e) Mounting bolt holes shall be round.
(f) Bolt hold diametrical clearance shall be 1/32 inch for 5/8 inch to 3/4 inch bolts and 1/16 inch for bolts over 3/4 inch.

3.5.3.2 Dimensions and weights. - The control equipment enclosures shall be as light and compact a possible, while consistent with specified performance and design. The maximum dimensions and weights of all bulkhead mounted cabinets shall be as specified in table IV.

3.5.3.3 Enclosure face. - When control arrangement A is furnished; all instruments, rheostats, circuit breakers, and other components needed for normal control of the equipment shall be mounted on the front of the panel above the centerline of the panel. The meters shall be mounted at the top of the panel.

3.5.3.4 Materials. - The cabinet structure may be either of steel or aluminum construction.

3.5.3.5 Insulating material. - In order to restrict the use of material which might catch fire or emit fumes when exposed to flame, the quantity of insulating material shall be kept to a minimum and used only where necessary to provide electrical insulation between live parts and to ground.

(h) Regulated voltage adjusting rheostat (see 3.2.12).
(i) Regulated frequency adjusting rheostat (see 3.2.16).
(j) Synchroscope 1/.
(k) Motor controller "start-stop" master switch.
(l) Circuit breaker 1/.
(m) Wattmeter for generator output 1/.
(n) Voltmeter transfer switch 2/.
(o) Voltage balance regulator 3/.

1/ Furnish with arrangement A only.
2/ Furnish with arrangement A, type II only.
3/ Furnish with type II.

3.5.2.2 Control equipment location - less than 30 kw. - For motor generators of less than 30 kw rating, items (a) through (d) shall be located either in an enclosure mounted atop the motor generator frame or in a bulkhead mounted, dripproof enclosure. Items (e) through (o) shall be mounted in a bulkhead mounted, dripproof enclosure. When mounted atop the motor generator, the equipment shall be cooled by the airstream of the motor generator.

3.5.2.3 Control equipment location - 30 kw and over. - For motor generator of 30 kw rating and over, items (a) through (d) shall be located in an enclosure mounted atop the motor generator frame. Equipment in this cabinet shall be cooled by the airstream of the motor generator. Items (e) through (n) shall be located in a bulkhead mounted dripproof enclosure. Item (o) shall be mounted in a separate, bulkhead mounted, dripproof enclosure.

3.5.2.4 Airborne noise. - The control equipment shall be designed to produce minimum airborne noise. The airborne noise limits, instrumentation and measurements shall be in accordance with MIL-STD-740 and the following:

Source: https://assist.dla.mil - Downloaded: 20260119T06:25Z
Check the source to verify that this is the current version before use.

3.5.3.6 Enclosure doors and removable covers. - Enclosure doors and removable covers shall be hinged on the left-hand side when facing front of the panel and shall be provided with knurled captive bolts with screw driver slot.

3.5.3.7 Positioning devices. - The hinged door shall be provided with stops and positioning devices to automatically stop it in its open positions and to prevent its moving from those positions due to gravity or ship motion. Its full open position shall be at least enough to permit unimpeded access and shall minimize obstruction of working or passage areas. Intermediate stops shall be provided when, added convenience for maintenance will result.

3.5.3.8 Grab rods. - All metered control panels shall be provided with vertical grab rod located so that an operator will have means of support while operating the panel. The rods shall be of smooth hardwood, or other non-conductive material supported with a sufficient number of brackets to provide adequate support for personnel under all operating conditions of the vessel. The rods shall not interfere with the opening of the doors nor shall their removal be required to open the door. The rods shall not interfere with operation of the equipment and shall not extend more than 6 inches from the front of the panel. The rod shall be capable of resisting a force of 400 pounds applied at any point on the rod without permanent distortion or damage to the rod or supporting means.

3.5.3.9 Operating and maintenance instruction. - A reduced size copy of abbreviated operating instructions, complete schematic diagram of the system and wiring diagram shall be laminated in plastic and stowed in the control equipment panels or enclosures. The diagram printing process shall be such that fading of the recorded information will not occur. The diagrams and instruction sheets shall be either mechanically attached to enclosure doors or stowed in racks in the enclosure in such a manner as not to interfere with operation of the equipment.

3.5.3.10 Ventilation. - Forced draft ventilation shall not be used except where the components are mounted in the intake airstream of the motor generator.

3.5.3.11 Accessibility. - All parts which require servicing, repair, replacement, or periodic adjustment during the life of the equipment shall be readily accessible through the front of the panel or enclosure. Accessibility shall be obtained by providing hinged full doors. Wherever practicable, subassemblies shall be utilized in the design of the equipment in order to reduce the time required for service and maintenance, for example, separate assemblies for the voltage detector and individual amplification stages. The use of plug in assemblies or quick-disconnect devices shall not be incorporated in the design.

3.5.3.12 Internal temperature. - The design of the equipment enclosure and the placement and selection of parts shall be such that no part will be exposed to a higher operating temperature than the maximum rated temperature for the part.

3.5.3.13 Heating. - Circuit design shall not require the use of heaters to obtain proper operation with varying ambient temperature or the variation in temperature rise within the enclosure due to heating of components under different operating conditions.

3.5.4 Component requirements. -

3.5.4.1 Component mounting. - In order to prevent the possibility of displacement of components due to shock and vibration, the use of slotted mounting bolt holes is prohibited. The holes shall be round with a maximum diametrical clearance of 1/16 inch over the mounting bolt diameter. Components previously approved by the command or agency concerned shall be mounted in a manner similar to the way they were mounted when subjected to the HI shock and vibration tests during their qualification testing.

3.5.4.2 Component and terminal markings. - All components, terminals and terminal boards shall be marked with the identifying designations shown on master drawings approved by the command or agency concerned. Wherever practicable, the designations shall be in accordance with MIL-STD-195. The identifying designations may be made by metal stamping, engraving, silk screening, stenciling, or rubber stamping, however, silk screening, stenciling, or rubber stamping may not be used on terminal boards.

3.5.4.3 Adjustable circuit elements. - The number of adjustable circuits shall be kept to an absolute minimum and limited to those adjustments required by maintenance personnel. When 100 percent adjustable resistors are used, a fixed series resistor shall be employed to protect circuit components from over-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

load due to improper adjustment of the variable resistor. A complete description and instructions for each adjustment shall be included in manuals.

3.5.4.4 Mechanical stability. - All controls of the equipment shall be so designed that when the equipment is properly installed and whether operating or nonoperating, the setting, position, or adjustment of any control shall not change due to normal service conditions.

3.5.4.5 Adjustment locking devices. - Adjustment locking devices, where used, shall be capable of retaining the controls in any given setting within the range of control under shock and vibration. The locking and unlocking action shall be easily and quickly accomplished with the equipment energized and shall not affect the setting of the control. When in the unlocked position, the locking device shall not interfere with the normal operation of the control. Where vernier controls are used, the locking device shall operate on both main and vernier controls.

3.5.4.6 End stops. - Each control which is not capable of an unlimited degree of motion without damage shall be provided with positive end stops to limit the degree of motion of the control to that within its normal excursion or within its range of calibration. Where flexible control shafts are employed, end stops shall be provided on the driving end of the shaft.

3.5.4.7 Motor generator starting shall be accomplished in accordance with the following:

(a) Applications requiring the use of size 4 controllers or less shall use controllers in accordance with MIL-C-2212.
(b) Applications requiring the use of size 5 controllers or greater shall use controllers in accordance with MIL-C-2212 or circuit breakers in accordance with MIL-C-17361.

3.5.4.7.1 Motor controllers shall be as required by the application, shall meet the requirements of MIL-C-2212 and the following:

(a) Operation shall be magnetic.
(b) Low voltage protection and overload protection with handrest shall be provided.

3.5.4.7.2 Circuit breakers shall be type AQB or ACB with electric closing attachment and under voltage protection in accordance with MIL-C-17361. The trip unit shall provide thermal protection required by MIL-C-2212. If such protection cannot be obtained with circuit breakers only, overload relays of MIL-C-2212 shall be used. Instantaneous trip shall have an "M" calibration.

3.5.4.7.3 Master switches shall be in accordance with MIL-C-2212.

3.5.4.8 Voltage regulator. - The voltage regulator and excitation system shall be designed to meet the system performance requirements specified in 3.2 when operating with its intended motor generator and as follows:

(a) Type - static or brushless. - That is, one which does not employ vibrating contacts, carbon piles, electron tubes or mechanically-moving parts such as relays and tachometers. This restriction does not apply to switching devices required to start the motor generator or to synchronize and interconnect the motor generators for parallel operation. Sliding contact variable resistors and rheostats other than those varied only by manual means will not be permitted. The required performance shall be obtained through circuits using transformers, reactors, magnetic amplifiers, silicon controlled rectifiers, diodes, transistors, rectifiers, resistors, capacitators and brushless rotating exciters. Schemes utilizing shunt reactive loading to obtain 1 percent voltage balance (type II) are permitted only if the shunt reactors are used to carry the unbalanced currents.
(b) Field flashing. - Field flashing shall be accomplished automatically from the rectified 60 cycle input power to the motor generator. No relays shall be used in the excitation system to provide generator field flashing upon starting.
(c) Compensation for parallel operation. - The reactive load division circuits provided to meet the parallel operation requirements specified in 3.2.23.1 shall be automatically energized as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

(1) Arrangement A -

a. A paralleling relay shall be energized automatically through auxiliary contacts on the generator circuit breaker so that generators being brought on will automatically be paralleled with the generator (or generators) already on the line when its circuit breaker is closed. The reactive load division circuits shall be adjustable to compensate for differences in reactive load sharing characteristics of the generators. The adjusting means shall be located on the inside of the regulator enclosure.

(2) Arrangement B -

a. A paralleling relay shall be provided in the voltage and frequency regulator cabinet for use in connection with a remote circuit breaker and shall operate as indicated in 3.5.4.8(c)(1)a.

(d) Overvoltage limitation shall be provided as a part of the voltage regulator to protect the equipment against overvoltage due to loss of sensing or failure of components in the voltage regulator. The overvoltage limiting circuit shall sense the generator terminal voltage and limit it by controlling the last stage of the exciter, should an overvoltage occur. The circuit shall limit the overvoltage 135 percent rated. The equipment shall not be damaged by overvoltage of 135 percent rated sustained for 15 minutes.

(e) Remote sensing. - When remote sensing is required (see 6.1), provision shall be made to automatically switch the sensing from the generator side of the circuit breaker (local sensing) to the remote sensing point when the circuit breaker is closed and power is available at the remote sensing point. When the generator circuit breaker is opened the sensing shall be automatically transferred back to local sensing. Auxiliary contacts on the generator circuit breaker or a relay may be used to transfer sensing. If a relay is used, the relay shall be deenergized for local sensing.

3.5.4.9 Frequency regulator. - The frequency regulator and excitation system shall be designed to meet the system performance requirements specified in 3.2 when operating with its intended motor generator and as follows:

(a) Type - Static type generator frequency sensing (see 3.5.4.8(a)).

(b) Circuit controlled - The specified performance requirements shall be met by either one of the following schemes:

(1) Control schemes which maintain frequency through completely static control of the input to an induction motor.

(2) Control schemes which maintain frequency through complete static control of the primary or secondary of a wound rotor motor.

(c) Compensation for parallel operation - Cross current compensation shall be provided to meet parallel operation requirements for real load division specified in 3.2.23.1. The method of energizing these circuits shall be as specified in 3.5.4.8(c). The real load division circuits shall be adjustable to compensate for differences in real load sharing characteristics of the motors. The adjusting means shall be located on the inside of the regulator enclosure.

(d) Where a maximum value of inrush current is specified (see 6.1), the current shall be limited by the design of the motor or by the circuitry of the frequency regulator.

(e) Effect on power source - When the motor generator and associated control equipment is connected to a 60 (+ 5 percent) cycle power source which has an impedance of: R= 1/_____ohms, X = 1/_____ohms, the following requirements shall be met at the input terminals:

(1) Power pulses produced on the system (sum of 3 phases) shall be at a frequency greater than 200 cycles per second. However, if power pulses in the range of 0 to 200 cycles per second are obtained, they may be permitted at the sole discretion of the procuring activity. Requests for such approval shall be made in writing and include: the frequency range of the pulses and the pulse power limit.

(2) The current modulation of the 60-cycle wave shall not exceed 10 percent of the full load current (see 3.1.9).

(3) The deviation factor of the 60-cycle voltage wave shall not exceed 5 percent (see 3.1.8).

1/

MG kva Input	60 Cycle source impedance R (Ohm)	X (Ohm
10	0.97	0.06
20	0.51	0.06
30	0.36	0.06

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the currentversion before use.

1/ MG kva Input	60 Cycle source impedance R (Ohm)	X (Ohm)
50	0.23	0.06
100	0.13	0.06
200	0.054	0.06
300	0.032	0.06
400	0.022	0.04
500	0.018	0.04
600	0.011	0.04

3.5.4.10 Relays. - Relays used for automatic connection of cross current compensation circuits shall be in accordance with MIL-C-2212.

3.5.4.11 Circuit breakers. - Circuit breakers, when required (see 3.5.2.1), shall be type ALB conforming to MIL-C-17588 for generator full load currents up to 100 amperes; or type AQB conforming to MIL-C-17361 for generator full load currents above 100 amperes. Circuit breakers shall be equipped with a 115-volt d.c. shunt trip device and an auxiliary switch.

3.5.4.12 Voltmeter. - The voltmeters, when required (see 3.5.2.1), shall be 4-1/2 inch, 250 degree scale switchboard type conforming to MIL-V-23151. The meter shall have a scale range of 430 to 470 volts and a center scale at 450 volts. The meter shall be suitable for operating directly from the generator output and not require potential transformers. A switch, in accordance with MIL-S-18396, shall be provided with the voltmeter so that all three phases may be monitored when type II is specified.

3.5.4.12.1 Voltmeter switches. - Voltmeter switches shall be in accordance with MIL-S-18396; shall be open with top, bottom, or back connections; shall be suitable for dead front mounting and shall have round (knurled) handles. Type JR switches (see drawings 9000-S6202-73075 and 9000-S6202-73295) may be used where the circuit voltage is 120 volts or less. Rating, number of poles and positions shall be as required to connect the voltmeter across each phase of type II motor generator sets and also across each line to neutral of 4-wire, type II motor generator sets.

3.5.4.13 Ammeter. - The ammeters, when required (see 3.5.2.1), shall be 4-1/2 inch, 250 degree scale switchboard type conforming to MIL-M-16034. The meter scale shall be the next size above full load generator output current and have a red line scale mark at rated amperes.

3.5.4.14 Frequency meter. - The frequency meters, when required (see 3.5.2.1), shall be 4-1/2 inch switchboard type. The meter shall conform to the applicable requirements of MIL-M-23167.

3.5.4.15 Synchroscope. - Synchroscopes, when required (see 3.5.2.1), shall be in accordance with MIL-I-16104.

3.5.4.16 Instrument transformers. - Instrument transformers shall conform to MIL-I-1361.

3.5.4.17 Wattmeter. - The wattmeters, when required (see 3.5.2.1), shall be 4-1/2 inch, 250 degree scale, 3-phase switchboard type conforming to MIL-W-19088. The wattmeter scale range shall be between 100 and 125 percent of the generator kw rating. The meter scale shall have a red line scale mark at rated kw.

3.5.5 Wiring and connections. -

3.5.5.1 Harnessing. - All wiring shall be neatly formed into groups which shall be laced with non-flammable (glass) cord or nylon ties and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock. The clamps shall be made of non-flammable material. Where metal is used, it shall be covered with a nonflammable material to prevent chafing of the wire insulation. There shall be no splices in the wire and all connections shall be made at the terminal studs of the devices or at terminal blocks. The wire groups running from the hinged doors shall be formed and clamped so that sharp bends do not occur with the panel in either the open or closed position.

3.5.5.2 Slack. - Sufficient slack shall be provided so as not to impair movement or put undue stresses on the wire or components in those places where movement of components may be expected. Slack shall also be provided to prevent undue stresses on terminal connections due to shock or vibration. When wire terminals

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

are used, sufficient slack shall be provided for at least two replacements of the part in the event that the wires are damaged or have to be clipped at the terminals during disassembly.

3.5.5.3 Cable entrance provisions. - Removable cable entrance plates shall be provided both top and bottom of each control panel so that the necessary cables can be brought into the enclosure. The top plate shall be provided with a gasket.

3.5.5.4 External cable connections. - External cable connections to the equipments shall terminate at a terminal board accessibly located near the bottom cable entrance and shall be accessible from the front of the enclosure. Where calibrated shunt leads are required, they shall terminate on the instrument studs. Excess length of calibrated shunt leads shall be neatly coiled and secured so as not to interfere with access to other equipment. Circuitous routing of wiring shall be avoided.

3.5.5.5 Insulation protection. - Wherever wires are run through holes in metal partitions, the holes shall be equipped with grommets for mechanical protection of insulation. Care shall be exercised in the running of wires to insure that they are not carried over or bent around any sharp corner or edge.

3.5.5.6 Studs. Studs of potted parts shall be so designed and fastened to the insulating strip or plate (or the enclosure itself if this conforms to insulation requirements) as not to cause any degradation in the moisture excluding property of the enclosure by the normal soldering and resoldering of external leads to the studs.

3.5.5.7 Wire end connectors. - The ends of each wire (except for pigtail leads or parts requiring solder connections) shall be connected to the part studs or to terminal boards by means of solderless connectors of a type approved by the command or agency concerned or by forming a wire around the part stud and retaining the wire loop in a cup or crimped washer. The strands of the wire loop shall be secured together by soldering. All nuts, bolts and screws used for electrical connections shall be secured by a locking device; however, a locking device need not be provided where solderless connectors for conductors below 4000 circular mils are used. Connections utilizing quick disconnect connectors (in accordance with MIL-C-5015) in conjunction with a positive locking arrangement, to prevent circuit opening under shock or vibration, may be used as approved by the command or agency concerned.

3.5.5.8 Soldering connections. - Pigtails (wireleads which are permanently affixed to a part), studs (soldering posts which are permanently fixed to a part), soldering lugs (projections which solder to ends of wires to facilitate attachments to solderless connections) and soldering terminals (items which are attached to the chassis, or insulating boards, strips, or posts to serve as solder junctions for two or more wires or leads) come under this heading. All soldering studs, lugs, and terminals shall be notched or otherwise provided with means for mechanically securing the wire or lead prior to the application of solder. All solder terminations shall be completely plated with tin or silver. All solder type studs and terminals shall be sturdily designed and shall be constructed of materials which will render them suitable for repeated soldering and unsoldering operations without breaking or loosening. All solder type studs shall be mounted in such manner as to preclude their loosening or rotation due to soldering operations or from strains due to attached wires or leads. No more than three wires, including wires from parts, shall terminate at any one terminal.

3.5.6 Identification and information plates. -

3.5.6.1 Identification plates. - Identification plates shall be furnished on each piece of control equipment which is separately mounted. When the control equipment is mounted in a common control panel, identification plates shall be furnished on the outside of the complete unit as well as on major individual components within the enclosure such as voltage regulators, motor controllers, circuit breakers, and so forth. The minimum data to be marked on the plate shall include the name of the unit, manufacturer's name, manufacturer's serial numbers, Federal stock number, drawing number, contract number, year of manufacture, rating data, a blank space for the Government inspector's official acceptance stamp and identification to associate the equipment with its parent equipment (such as, for use with 10 kw, 440 VAC/450 AC MG SNSN).

3.5.6.2 Information plates. - The function of each control and indicator on the face of the panel as well as the function of each internal adjustment, connector and test point, within the enclosure shall be indicated by means of information plates adjacent to the device.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

3.6 Repair parts. - Onboard repair parts, based on the total number of identical parts or sets of parts furnished for each ship on a contract or order, shall be supplied as specified in table IX. This table includes parts for the complete motor generator set, including auxiliary devices such as the voltage regulator, motor starter, and so forth, and takes precedence over repair part requirements in individual referenced specifications.

Table IX - Quantities of onboard repair parts to be furnished per ship.

Parts required	Identical parts, or sets[1/] of parts, installed per ship							
	1	2	3	4	5-8	9-20	21-50	50-100
Bearings, sets	1	1	1	1	2	3	4	5
Brush holders, complete with springs	1	1	1	1	2	3	4	5
Brush holder springs	1	2	3	3	4	6	7	8
Brush rigging insulation, sets	1	1	1	1	2	3	4	5
Brushes, sets	2	3	4	4	5	6	8	12
Capacitors ea. size and type	1	1	1	1	2	3	4	5
Circuit breaker trip element if AQB	1	1	1	1	2	3	4	5
Type breaker is furnished, or complete circuit breaker if ALB type is furnished								
Contacts for rheostats, motor controller (except for overload devices) and relays sets	1	1	1	1	2	3	4	5
Overload heaters for motor controller, sets	1	1	1	1	2	3	4	5
Rectifiers, ea. size and type	1	2	3	3	4	6	7	8
Resistors, ea. size and type	1	1	1	1	2	3	4	5
Rheostats, ea. size and type	1	1	1	1	2	3	4	5
Saturable reactors only if the reactors contain magnet wire of less than 1000 Circular mils cross-sectional area Complete set (or four, whichever is less)	1	1	1	1	1	2	3	4
Shunt coils for motor controller and relays	1	1	1	1	1	2	3	4
Springs for motor controller (except for overload devices), sets	1	1	1	1	2	3	4	5
Switches, ea. size and type (complete)	1	1	1	1	2	3	4	5
Transistors, ea. size and type	1	2	3	3	4	6	7	8
Voltage regulator diodes	1	2	3	3	4	6	7	8
Voltage reference diodes	1	2	3	3	4	6	7	8
Fuses	1	2	3	3	4	6	7	8

1/ A set is all parts of the same size and type installed in one motor generator and its associated controls.

3.6.1 Irreplaceable parts. - If any part listed in table IX forms part of an assembly or subassembly and is not normally a replaceable part of the assembly, then the complete assembly shall be furnished in lieu thereof as a repair part.

3.6.2 Matched parts. - Where matching of parts is required in the equipment, the matched set of parts shall be considered one part and furnished as such if required as a repair part.

3.7 Drawings. -

3.7.1 General requirement. - Drawings shall be in accordance with MIL-D-1000 and as specified herein.

3.7.2 Kind of drawings. - Drawings shall be type II in accordance with MIL-D-1000/2 and the following:

(a) Preliminary drawings.
(b) Final drawings.

3.7.2.1 Preliminary drawings. - These drawings which shall be submitted for approval shall be either blue prints, blue line prints, or equal, and shall contain the information required by 3.7.3.3, 3.7.3.4 and 3.7.3.5

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

3.7.2.2 Final drawings. - Final drawings shall be either microfilms, tracings, or direct reading blackline prints of the approved preliminary drawing in accordance with MIL-D-1000/2 and MIL-M-9868/1. Where microfilms are furnished, one full size nonreproducible copy of each drawing shall be furnished.

3.7.3 Drawings required. - The following drawings are required to be furnished:

(a) Motor-generator drawing (3 sheets maximum).
(b) Control panel drawing exclusive of motor controller and circuit breaker (5 sheets maximum).
(c) Motor starter drawing (as required).
(d) Voltage and frequency regulator drawing (5 sheets maximum).
(e) Voltage balance regulator (3 sheets maximum).
(f) Installation drawing (1 sheet maximum).

3.7.3.1 When separately mounted control equipment is furnished in lieu of a complete control panel (see 3.5.2.2 and 3.5.2.3), components which may be necessary to make complete system but are not furnished by the contractor (such as circuit breakers, switches and so forth), shall be indicated on the installation drawing and shown in the schematic and wiring diagrams and a note added to indicate that they were not furnished by the contractor.

3.7.3.2 When a complete control panel is furnished, the regulator drawings shall be counted as part of the five sheets required for the control panel.

3.7.3.3 Motor-generator drawing. - Information to be shown on the motor-generator master drawing shall be as follows:

(a) Descriptive data. -
(1) Identification
a. Manufacturer's type, class or model number.
(2) Reference specification
a. A statement as follows: "The a.c./a.c. motor-generator shown hereon conforms to MIL-M-19633, dated_____with the following exceptions: (State none or list all exceptions as applicable)."
(3) Guaranteed performance
a. Overall efficiences at 1/4, 1/2, 3/4 and full load.
(4) Motor generator classification
a. Enclosure.
b. Speed.
c. Ambient temperature.
d. Mounting.
e. Duty.
f. Insulation class.
(5) Motor classification
a. Horsepower, frame size, voltage, speed, full load current, phase and frequency.
b. Stator core diameter, core length and nominal air gap.
c. Rotor core diameter, core length.
d. Brush information consisting of manufacturer's grade, Military grade, thickness, width, length, top and bottom bevel, number of brushes per stud and brush pressure.
e. Rotor current and voltage.
(6) Generator classification
a. Kw rating, voltage, phase, power factor, frequency current, excitation current and voltage (no load and full load) and insulation class.
b. Generator constants.
1. Transient reactance (maximum calculated value only).
2. Subtransient reactance (minimum calculated value only).
3. Open-circuit transient time constant (calculated value).
4. Synchronous reactance (calculated value).
5. Negative sequence reactance.
c. Stator core diameter, core length, nominal air gap.
d. Rotor core diameter, core length.
e. Number of collector rings, ring diameter, ring material, wearing depth, insulation to ground, creepage to ground.

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the current version before use.

f. Brush information consisting of manufacturer's grade, Military grade, thickness, width, length, top and bottom bevel, number of brushes per stud and brush pressure.
g. Identification of slipring resurfacing tool (manufacturer and type) when slipring are used.

(7) Shaft, bearings and lubrication
a. Shaft material.
b. Type of bearings.
c. Type of lubrication.
d. Note identifying the type and class of the applicable Military standard bearing puller and adapter (see GGG-P-781).

(8) Weights
a. Weight of complete motor generator.
b. Weight of rotor.

(b) Motor winding data. -
(1) Stator
a. Winding data of stator, including:
Number of poles.
Type of connection.
Number of slots.
Number of coils.
Winding pitch in slots.
Turns in series per coil.
Conductor circular mils or size, form (round or rectangular), and developed length.
Conductor insulation and specification.
Resistance between terminal in ohms @ 25°C.
Weight of copper in pounds.
b. Insulation materials and their dimensions and applicable specifications of the following:
Slot cell.
Spacer.
Top wedge.
"U" wedge (where used).
Insulation between phases.
Insulation on coil extension.
Insulation on coil leads and connections.
Complete stator, specification and grade of varnish.
Lead wire.
c. Cross sectional view of slot showing details of slot wedges, slot armor, coil spacers and slot tubes.
d. A schematic wiring diagram.
e. Table of all insulation material indicating type of material, form, thickness, applicable specification and indicate where used. Materials listed in this table need not be repeated in the basic list of materials.

(2) Rotor
a. Squirrel cage - Number of bars, bar material and size, size of end ring and material of end ring.
b. Wound rotor - data as specified in 3.7.3.3(b)(1).

(c) Generator winding data. -
(1) Armature winding data to include type of winding, number of slots, number of coils per slot, turns per coil, pitch of coil, size of conductor insulation (specification and type) feet of wire per coil, weight of copper per coil, resistance of wound armature in ohms at 25°C.
(2) Armature, winding diagram, facing connection end.
(3) Cross sectional view of armature slot showing coils and all insulation.
(4) Cross sectional view of field poles showing coils and all insulations.
(5) Dimensional plan and end view of each type of armature and field coils. Dimensions before and after forming shall be shown.
(6) Field winding data to include size and shape of conductors, conductor insulation (specification and type), turns per coil, feet of wire per coil, number of conductors in parallel and resistance per coil in ohms at 25°C.
(7) Field coil connection diagram.
(8) Table of all insulation material indicating type of material form, thickness, applicable specification and indicate where used. Materials listed in this table need not be repeated in the basic list of materials.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the currentversion before use.

(d) The following additional insulation data shall either be shown on each motor-generator master drawing or may be shown on a separate insulation procedure master drawing showing the various processes, procedures and methods used by the manufacturer. If the latter is selected, the motor-generator master drawing shall reference the applicable portions of the insulation procedure master drawing.
(1) Sequence of all winding and insulating operations and applicable specifications for all material used.
(2) Tape application including type, number of layers, amount of overlap, and so forth.
(3) Sketches as necessary to completely describe all operations.
(4) Ground insulation, slot cell insulation, phase insulation, coil spacers and separators, end turn insulation, insulation under banding wire, support pads, distance that slot cell insulation extends beyond laminations, alternate materials slot wedges, method of bringing out and insulating leads from each coil, method of securing windings, and so forth.
(5) Varnish treatment schedules including type of varnish, impregnation methods, time cycles, and baking temperatures for wound coils prior to installation as well as treatments for completed subassemblies.

(e) Test reports. -
(1) The file number and date of shock test and of the approval letter shall be referenced on the drawing.
(2) The periodic test report number or numbers and the file number and date of the approval letter shall be referenced on the drawing.

(f) Construction details. -
(1) An assembly drawing showing a side and two end views with all parts of the motor generator identified with piece numbers in the list of materials. The side view shall show a longitudinal section of the motor-generator above the centerline of the shaft. The end views shall show quarter sections (or more if necessary) of the motor and generator. These assembly views shall be drawn so that they will be legible when the drawing is reduced to manual size. The assembly views shall show all information as indicated herein. Separate detail views of subassemblies shall be provided where the sectional views do not clearly show the information required.
(2) Method of securing all rotating elements to the shaft. This includes bearing inner races, fans, collector rings, armatures, rotating field assemblies, spiders, spacers, sleeves, and so forth. The drawing shall clearly indicate the method used to prevent both axial and rotational movement of all items secured to the shaft. Where interference fits are used to prevent axial movement, the inner and outer diameters and tolerance, shall be indicated.
(3) Method of securing all stationary elements to the frame. Details furnished to be the same as indicated for 3.7.3.3(f)(2).
(4) Bearing housing construction, fits and tolerances. Bearing fits and tolerances may be indicated in tabular form.
(5) Baffles and screen guards and method used to attach them to the end brackets or frames.
(6) Method of attaching terminal box to frame.
(7) Method of bringing cables out of frame into terminal box and method of protecting these cables against chafing at sharp metallic corners.
(8) Method of securing the motor and generator leads to the terminal boards in the terminal box.
(9) Overall dimensions of terminal box, and the number and size of cover bolts.
(10) A detailed working drawing of the shaft including dimensions, finishes and tolerances.
(11) Sectional view of collector ring assembly showing direction of rotation, design of brush holders and brush rigging, including studs, and method of supporting the brush holder rigging, including studs, against excessive deflection due to high impact shock. The drawing of the brush holder shall show the basic design, method of adjusting spring pressure and method of attaching brush shunt. The materials of the brush holder body, brush holder spring, and brush holder rigging shall be indicated in the list of material.
(12) Direction of air flow through the frame.
(13) Lifting means provided.
(14) All overall dimensions including mounting and maintenance clearances.

3.7.3.4 Control equipment drawing. - The control panel, voltage regulator, voltage balance regulator and frequency regulator master drawings shall include the following information, as applicable. The motor controller drawings shall include information as required by MIL-C-2212 and MIL-R-15109, respectively. The manufacturer of all purchased items shall be indicated on the drawing.

(a) Descriptive data for the equipment. -
(1) Type of equipment

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

a. Control panel, voltage regulator, voltage balance regulator, frequency regulator.

(2) Manufacturer's identification
a. Type, class, model number.

(3) Reference specification
a. A statement that the equipment conforms to MIL-M-19633, dated_____with the following exceptions: (State none or list exceptions.)

(4) Equipment classification (as applicable)
a. Type of equipment.
b. Enclosure.
c. Shock classification.
d. Ambient temperature.
e. Electrical ratings, voltage, current phase frequency, voltage, and so forth, as applicable.
f. Mounting.
g. Ventilation.
h. Weight of complete assembly or part where furnished separately.
i. Insulation class.
j. Special features.

(5) Test report
a. The file number and date of shock test and of the approval letter shall be referenced on the drawing.
b. The periodic test report number or numbers and the file number and date of approval letter shall be referenced on the drawing.

(b) Parts design data. -

(1) Transformers, reactors and magnetic amplifiers
a. Core material.
b. Winding data including the number of turns, tape, wire size and type, terminal identification, d.c. resistance between terminals at 25°C, nominal voltage between terminals, volt - amp. ratings for transformers, and current ratios for current transformers.
c. Part designation as assigned in the elementary schematic diagram and part number used in the list of material.
d. Complete material details including applicable specifications for materials used. Details of core boxes, layer and coil insulation, ground insulation, method of impregnation and treatment, potting methods and materials, terminal boards, grommets and leads shall be included.

(2) Resistors
Style, resistance, voltage rating, wattage, type designation or part number, type of mounting, type of terminals, type of adjusting band, and any other special data which may be necessary for ordering exact duplicates from the resistor manufacturer.

(3) Rheostats
Nominal diameter, resistance, wattage, voltage rating, current rating, number of steps and part number.

(4) Capacitors
Style, capacitance, voltage rating, type designation or part number. Also indicate the maximum a.c. (peak and rms) and d.c. (average) voltages which will be applied to the capacitor in the application and the condition at which these maximum voltages occur (transient load conditions, 50 percent rated current, and so forth). If dry electrolytic capacitors are used, add the following drawing note:
"It is to be noted that these electrolytic capacitors deform and may become unstable. Idle storage or shelf life should not exceed four years."

(5) Semiconductor devices.
Each device shall be identified in the list of material by manufacturer's part or drawing number to permit direct ordering (JEDEC number is adequate). Also provide mounting details and application (not device capability) data as follows:
a. Rectifiers using silicone diodes.
1. Circuit symbols and piece number.
2. Maximum reverse voltage peak and rms (complete rectifier circuit).
3. Maximum load current d.c. amperes (complete circuit).
4. Maximum diode case temperature at design ambient-use for diodes rated against core temperature.
5. Maximum ambient temperature directly adjacent to diodes-use for diodes rated against ambient temperature - no cooling plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

6. Cooling plate size, thickness and material where so mounted.
7. Air velocity if forced cooled (linear feet per minute.)
8. Circuit (for example, single phase bridge, and so forth).
9. Total diodes in circuit.
10. Diodes in series per arm.
11. Diodes in parallel per arm.
12. Description of insulation hardware if insulated from a cooling plate.

b. Voltage regulator diodes.
1. Circuit symbol(s) and piece number.
2. Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes and still give required equipment performance _____volts to _____volts.
3. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance) ± _____mv.
4. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds):_____ma.
5. Maximum transient reverse current under any operating condition _____ma for _____sec duration and_____frequency of occurrence.
6. Maximum ambient temperature at diode (no cooling plate).
7. Maximum case temperature of diode (complete for diodes rated against case temperature.)
8. Cooling plate size, thickness and material where used.
9. Air velocity if forced cooled (linear feet per minute).
10. Total diodes in circuit.
11. Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit).
12. Description of insulation hardware if insulated from a cooling plate.

c. Voltage reference diodes (See regulator diodes also).
1. Circuit symbol(s) and piece number.
2. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s):_____volts to _____volts.
3. Error permitted in an installed voltage reference diode under all conditions of operation: ± _____millivolts.
4. Circuit design current through voltage reference.
   (a) Nominal or average value:____ma.
   (b) Regulation about nominal value: ±_____ma.
5. Maximum ambient temperature (include internal heat rise above outside design ambient):_____a.
6. Number of voltage reference elements in series:_____(operation in parallel is not permitted).

d. Transistors.
1. Circuit symbols(s) and piece number.
2. Max. collector to emitter voltage_____volts.
3. Max. collector to base voltage_____volts.
4. Max. collector current_____ma.
5. Max. base current_____ma.
6. Maximum collector power dissipation_____mw.
7. Maximum ambient temperature (include internal heat rise at transistor at 50°C. design ambient_____°C.
8. Maximum core temperature of transistor at 50°C. equipment design ambient and any condition of required equipment operation_____°C. (Where rating is based on case temp.).
9. Air velocity of forced cooled.
10. Cooling plate size, thickness and material where so mounted.
11. Description of insulating hardware if insulated from a cooling plate.

e. Selenium surge suppressors.
1. Circuit symbol(s) and piece number.
2. Supply voltage, steady state rms (nominal and maximum).
3. Peak discharge current.
4. Peak and average power dissipation.

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the currentversion before use.

5. Maximum cell temperature.
6. Peak surge voltage with suppressor(s) applied.
7. Circuit arrangement and terminal marking.

(6) Meters.
Scale, range and red line at rated marking.

(7) Circuit breakers.
Type, frame size, thermal element rating, instantaneous trip calibration, and any special features which may have been provided.

(8) Motor controllers.
Type, size, protection features, overload heater rating, and any other special features which may have been provided or minor deviations from the master drawing for the motor starter.

(c) Schematic diagram. -
(1) A schematic diagram of the complete system to clearly represent the operation of the equipment. The following features shall be incorporated:
a. Each part, such as resistors, capacitors, and so forth shall be identified by symbol (R-resistor, T-transformer, L-reactor, and so forth). Each symbol shall be used only one time in the complete power supply.
b. In addition to the part designation, the numerical value of all resistances, capacitances and inductances shall be noted adjacent to the part.
c. Identification of all terminals shall be shown.
d. A brief description of theory of operation of the system to enable the approval authority to understand the circuitry.
e. Simplicity in reading and understanding the operation of the system shall be stressed throughout.
f. The main power lines should be drawn heavier than the control lines for ease in tracing power circuit.

(d) Wiring diagram. -
(1) A wiring diagram for each of the major subassemblies and interconnection of the subassemblies. The following features shall be incorporated in the drawing:
a. All parts shall be shown in correct relation to physical location within a particular enclosure.
b. Terminals of all assemblies and subassemblies shall be clearly shown and marked.
c. Designation of parts shall conform to those assigned in the schematic diagram. Also, the piece numbers corresponding to those shown in the list of material shall be indicated.
d. Wire size, type, applicable specifications, and type of terminals shall be shown in tabular form.
e. Lead identification.
f. Continuous lines to represent leads between components need not be drawn if the lead identification used clearly indicates the termination point of each lead.

(e) Construction details. -
(1) Two representative views of each enclosure showing the following:
a. Overall dimensions of the enclosure.
b. Drilling and mounting dimensions.
c. Outline of all parts, subassemblies and assemblies contained in each enclosure identified with item or piece number corresponding to item or piece number contained in the list of material.
d. The parts need not be drawn in detail nor to scale but should represent general configuration of the part and show its general location in the enclosure.
e. Details of construction showing cable entrance provisions; door stop and positioning device; grab rods; captive bolts; gaskets; and other details as applicable.
f. Outline drawing of parts or components furnished separately (not mounted in an enclosure) to show critical dimensions including mounting dimensions.

3.7.3.5 Installation drawings. - Data to be shown on these drawings shall be as follows:
(a) Dimensional outline of all major assemblies (motor generator, control panel, and so forth), showing overall and principle dimensions in sufficient detail to establish the limits of space in all directions required for installation, operation and servicing, exclusive of space required for personnel. Include the amount of clearance required to permit opening doors and hinged panels; clearances required for withdrawal of parts and subassemblies such as end

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

brackets, rotating assemblies, and so forth, for the motor-generator and chassis for the control equipment panels; clearances for cable access, and clearances for ventilation. Also indicate the location of meters and operating controls on control equipment panels so that the panels may be located at a convenient operating height.

(b) A table of reference drawings to include drawing number of each major assembly, weight of each enclosure and separately furnished assembly and heat dissipation of major assemblies.
(c) Center of gravity of all major assemblies.
(d) A block diagram of all major assemblies showing interconnection between these assemblies. Terminal identification, as well as maximum current rating for each cable and maximum resistance for control cables should be shown.
(e) Maximum heat dissipation of all major assemblies.
(f) Motor starting current and maximum time to attain rated speed.
(g) Grade of mounting bolts to be used.
(h) Input data: voltage, frequency, full load current, number of phases, wattage.
(i) Output data: voltage, frequency, full load current, number of phases, wattage.
(j) Rated speed.
(k) Enclosure: ambient temperature and insulation class.
(l) Any special instructions for handling and installing.

3.8 Manuals. - The dimensions of the manual shall be 11-5/8 inches high and 8-3/4 inches wide. Manuals shall conform to type II of MIL-M-15071 and shall contain the following information in the order listed.

(a) Front material
  (1) Cover
  (2) Approval and procurement record page
  (3) List of effective pages
  (4) Table of contents
  (5) List of figures
  (6) List of tables
  (7) List of drawings
(b) Chapter 1 - General information
  (1) Section 1 - General data
    a. Table of major components with dimensions and weights indicated.
    b. Power supply requirements (voltage, frequency, current, phase permitted voltage and frequency variation, and so forth).
  (2) Section 2 - Introduction
    a. Brief description of equipment furnished and its basic parameters. Regulation, voltage balance, transient response, parallel operation, and so forth.
    b. Block diagram of complete system.
(c) Chapter 2 - Detailed description
  (1) Section 1 - Motor generator
    a. Detailed description of major parts or assemblies.
    b. Photograph.
  (2) Section 2 - Control equipment
    a. Detailed description of major parts or assemblies and voltage regulator, frequency regulator, voltage balance, and so forth.
    b. Photographs of control equipment with doors open and closed with major parts identified by arrow and symbol.
(d) Chapter 3 - Principle of operation
  (1) Section 1 - Voltage regulator
    a. Schematic diagrams of various assemblies of the voltage regulator with explanation of function of each component.
  (2) Section 2 - Frequency regulator
    a. Schematic diagram of each assembly of the frequency regulator with explanation of function of each component.
  (3) Section 3 - Voltage balance regulator
    a. Schematic diagram of various assemblies of the voltage balancer with explanation of function of each component.
(e) Chapter 4 - Installation
  (1) Section 1 - Installation instruction
  (2) Section 2 - Recommended procedure and adjustments for initial starting

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the currentversion before use.

(f) Chapter 5 - Operating instruction
  (1) Section 1 - Start up
  (2) Section 2 - Automatic operation
  (3) Section 3 - Parallel operation
  (4) Section 4 - Shut down
(g) Chapter 6 - Maintenance
  (1) Section 1 - Safety precautions
  (2) Section 2 - Preventive maintenance
    Routine inspection and care of equipment
  (3) Section 3 - Corrective maintenance
    List of instruments, special tools, recording devices and hardware needed
    Open loop test
    Trouble shoot charts
    Tables showing points of measurements of voltage, current, resistance and so forth
    Adjustments after repair
(h) Chapter 7 - Parts identification
  (1) Section 1 - Introduction
  (2) Section 2 - Parts tabulation
(i) Chapter 8 - Test data
  (1) Section 1 - Introduction
  (2) Section 2 -
    a. Voltage and frequency regulation, voltage balance, parallel operation, harmonic content, no load saturation curve, motor generator winding resistances at 25°C., insulation resistance (cold), locked rotor voltage, current and watts, fault current, curves drawn from oscillograms showing voltage and frequency transient response (dip and rise) as a function of time, deviation factor, full load heat run.
(j) Chapter 9 - Drawings - Reduced size prints of approved motor generator, control equipment and installation master drawings.
(k) Chapter 10 - Appendix
  (1) Section 1 - Motor controller insert
  (2) Section 2 - Circuit breaker insert.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality program requirements. - The supplier shall provide and maintain a quality program acceptable to the Government. The program shall be in accordance with MIL-Q-9858 and as specified in the Appendix to this specification. The supplier is in no way relieved of the final responsibility to furnish motor generators meeting the requirements of this specification.

4.3 Sampling. -

4.3.1 Sampling procedures for routine tests. - All motor generators on the contract or order shall be subjected to the routine tests specified in table X. The tests shown therein shall be performed on completely assembled units. Nonconforming equipment shall be individually rejected.

4.3.2 Sampling procedure for periodic tests. - The first motor generator power supply of each design and size offered for delivery shall be subjected to the periodic tests specified in table X. Thereafter, one periodic test on identical equipment will be required every 12 months. If a motor generator sample does not meet performance requirements under periodic tests, that motor generator and those of the same design and lot shall be rejected. After the design has been modified and the revisions incorporated into the sample, the sample may be resubmitted for tests. A periodic test shall be required after any change in design which affects the performance characteristics. If routine test data reveals variations beyond a normal manufacturing tolerance, the inspector may require that any or all of the periodic tests be made on a particular motor generator to demonstrate that it conforms to this specification.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

Table X - Routine and periodic tests.

Description of test	Applicable test paragraph		Test to be conducted while motor-generators and control equipment are operating together as a complete system	Test which may be conducted on motor-generator alone	Test which may be conducted on control equipment alone
	Routine tests	Periodic tests			
Creepage and clearance	4.6.1	4.6.1		X	X
Enclosure	4.6.2	4.6.2		X	X
Weight		4.6.3		X	X
Resistance at 25°C.	4.6.4	4.6.4		X	
Insulation resistance (cold)	4.6.5	4.6.5		X	X
Dielectric	4.6.6	4.6.6		X	X
Airborne noise	4.6.7.2	4.6.7		X	
End play	4.6.8	4.6.8		X	
Internally excited vibration:					
Motor generator	4.6.10	4.6.10		X	
No load saturation curve for generator		4.6.11		X	
Synchronous impedance curve		4.6.12		X	
Shock		4.6.13	X		
Regulated voltage adjustment range and steps		4.6.14	X		
Regulated frequency adjustment range and steps		4.6.15	X		
Automatic voltage and frequency regulation	4.6.16.2	4.6.16.1	X		
Transient response of voltage regulator (deviation and recovery time)	4.6.18	4.6.18	X		
Transient response of frequency regulator (deviation and recovery time)	4.6.19	4.6.19	X		
Parallel operation (kw and kvar sharing)		4.6.20	X		
Short circuit characteristics		4.6.21	X		
Rated load heating		4.6.22	X		
Resistance (hot)		4.6.4		X	
Insulation resistance (hot)		4.6.23	X		
Efficiency at 4/4, 3/4, 2/4 and 1/4 load		4.6.24	X		
Harmonic analysis		4.6.25	X		
Amplitude modulation		4.6.26	X		
Deviation factor		4.6.27	X		
Line unbalance		4.6.28	X		
Voltage balance		4.6.29	X		
Overvoltage limitation		4.6.30	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Table X - Routine and periodic tests (cont'd.).

Description of test	Applicable test paragraph		Test to be conducted while motor-generators and control equipment are operating together as a complete system	Test which may be conducted on motor-generator alone	Test which may be conducted on control equipment alone
	Routine tests	Periodic tests			
Effect on power supply		4.6.36	X		
Inclined operation		4.6.9		X	
Starting tests		4.6.31		X	
Component ratings	4.6.32			X	
Repair parts		4.6.33			
Electrical balance of motor		4.6.35		X	

4.4 Test reports and approval.-

4.4.1 Test results shall be recorded by the supplier on test forms prepared by the supplier and approved by the Government inspector. The test data shall be bound into 9 by 11-1/2 inch binders. Each test report shall be identified by the number of this specification and by sufficient information to identify the equipment. No special format will be required; however, a table of contents shall be included at the beginning of the report and list each test required by table X.

4.4.2 Each test shall be prefaced by the name of the test and applicable test paragraph as indicated in table X. The data submitted shall be copies of the actual data taken on the test floor and not retyped data. The forms used shall allow sufficient columns to make instrument corrections and necessary calculations. When oscillograms are required, each trace shall be identified as to the circuit measurement involved and its calibration shall be indicated to permit ready use without reference to associated documents.

4.4.3 All test data shall be approved by the cognizant inspector prior to the shipment of the equipment. Upon approval, one file copy of all periodic test data shall be forwarded to the command or agency concerned.

4.5 Special provisions.-

4.5.1 Periodic tests.- The shock test and weight measurement need only be conducted on the first unit of a given design.

4.5.2 Routine tests.- Routine tests which require that the motor generator and controls be operated together may be conducted as follows:

(a) Use one motor generator to test all control units.
(b) Use one set of controls to test all motor generators.
(c) Test any motor generator and any set of controls together.

The motor generator and control which is used for the complete periodic test shall not be used in the routine testing of any of the other units.

4.6 Test methods.-

4.6.1 Creepage and clearance.- Creepage and clearance distances between current carrying parts in the motor generator and control equipment shall be measured to determine compliance with MIL-E-917.

4.6.2 Enclosure.- Acceptability of the enclosure shall be determined by visual examination of measurements to determine compliance with 3.3.5.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

4.6.3 Weight. - The weight of the complete motor generator, complete rotating element of the motor generator, complete control panel (or each individual control equipment enclosure and switchboard mounted components in the case of separately mounted control equipment) shall be measured and recorded.

4.6.4 Resistance (cold and hot). - The resistance of all motor and generator windings shall be measured when the equipment is at same ambient temperature and both resistance and ambient shall be recorded. The resistance of the windings measured after the rated load heat run (see 4.6.22.2) and the calculated temperature for this resistance shall be recorded adjacent to the cold resistances on the test report.

4.6.5 Insulation resistance (cold). - This test shall be made before the dielectric tests. Prior to the application of the test voltage, all circuits shall be thoroughly discharged by shorting them to the frame and the motor generator may be disconnected from the control equipment. The test voltage shall be applied between all electrically isolated circuits (motor generator as well as control equipment) and between each circuit and ground. Circuit diagrams should be carefully studied prior to conducting this test to ascertain that circuits which may be isolated by transformer are not inadvertently neglected. When testing between the circuits and ground, all circuits of the control equipment and the motor generator may be tied together so that only one test voltage need be applied to each, the motor generator and the control equipment providing that the insulation resistance when tested in this manner shall be as specified in 3.2.5. Insulation resistance shall be measured with an insulation resistance indicating instrument. The time of test voltage application shall be not less than 60 seconds. The temperature of the circuits at the time of the test shall be measured and recorded. Insulation resistance measurements shall be corrected to 25°C. Unless otherwise specified in the contract or order, corrections shall be made on the basis of insulation resistance doubling for each 15°C. decrease in temperature. The relative humidity at the time of the test shall be measured and recorded.

4.6.6 Dielectric test. -

4.6.6.1 The dielectric test shall be made after all other tests have been completed. The dielectric test shall be made on the completely assembled motor generator and control equipment and not upon individual parts within the motor generator or control equipment. An exception is made in the case of repair parts which require dielectric tests.

4.6.6.2 Points of application. - The test voltage of a magnitude as specified in 3.2.24 shall be successively applied between each electric circuit and all other electric circuits and metal parts. No component, however, in an assembled unit shall be subjected to a dielectric test voltage higher than that test voltage required for that component. When the unit test voltage is higher than that required for an individual component, that component shall be disconnected from the circuit. The test voltage shall be applied in such a manner as to preclude the possibility of pitting the bearings in case of insulation failure. Voltage need not be applied between stationary and rotating windings of the motor generator.

4.6.7 Airborne noise. -

4.6.7.1 By instrument (for periodic tests). - Airborne noise measurements shall be made in accordance with MIL-STD-740 with the equipment operating at rated load and nominal input voltage to determine compliance with 3.4.4.

4.6.7.2 By observation (for routine tests). - Observation shall be made of equipment noise during the progress of testing to determine that the equipment is free from any degree of noise comparably greater than that inherent in the given type and size of unit. If any objectionable noise is detected the inspector may require a test by instrument as specified in 4.6.7.1 to ascertain that the unit meets the requirements of 3.4.4.

4.6.8 End play. - The end play of all motor generator rotating elements shall be measured to determine compliance with 3.4.6.5.

4.6.9 Inclined operation. - The motor generator and control equipment shall be tested for inclined operation. The motor generator and control equipment shall be run at no load for a period of not less than 30 minutes at each of the positions of inclination as specified herein. The control equipment may control the motor generator during the test or be tested separately under simulated no load loading. During the progress of these tests it shall be ascertained that there is no apparent overheating of any component, dropping out of relays, circuit breakers, and so forth and the mechanical balance is as good as it was in the normal horizontal

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the current version before use.

position, that there is no pounding or grinding at the bearings, and that the lubrication is satisfactory. The angle of inclination shall be as follows:

(a) Motor generator.-
(1) Shaft inclined 15 degrees, generator end low.
(2) Shaft inclined 15 degrees, motor end low.
(b) Control equipment.- Only control equipment continuing contactors, relays, circuit breakers or other devices employing a movable arm shall be tested.
(1) Inclined 15 degrees in any direction.

4.6.10 Internally excited vibration test.- Each motor generator shall be tested to determine compliance with 3.4.5.2. This test shall be conducted after the motor generator has been operated at rated load. After balancing, the unit shall be completely assembled and, mounted elastically at a natural frequency less than one-quarter of the rotational frequency of the unit. To accomplish this, the minimum static deflection of the mounting should be determined as shown on figure 1. In no case shall the deflection exceed one-half the original height of the elastic element. Measurements shall be made on the bearing housing and in the direction giving the maximum vibration when the motor generator is operating at rated load. Test data submitted for each machine shall include a description of the test set up including compression of elastic mount, instruments used, location of readings taken, actual instrument readings, and any corrections or calculations made. This test shall also be run as part of the post shock tests of the unit which was HI shock tested in order to ascertain that the HI shock test has not affected the unit in such a manner that the equipment does not meet the post shock internally excited vibration limits of 4.6.13(b)(6). Failure to meet the internally excited vibration limit after shock shall be considered as cause for rejection of the equipment.

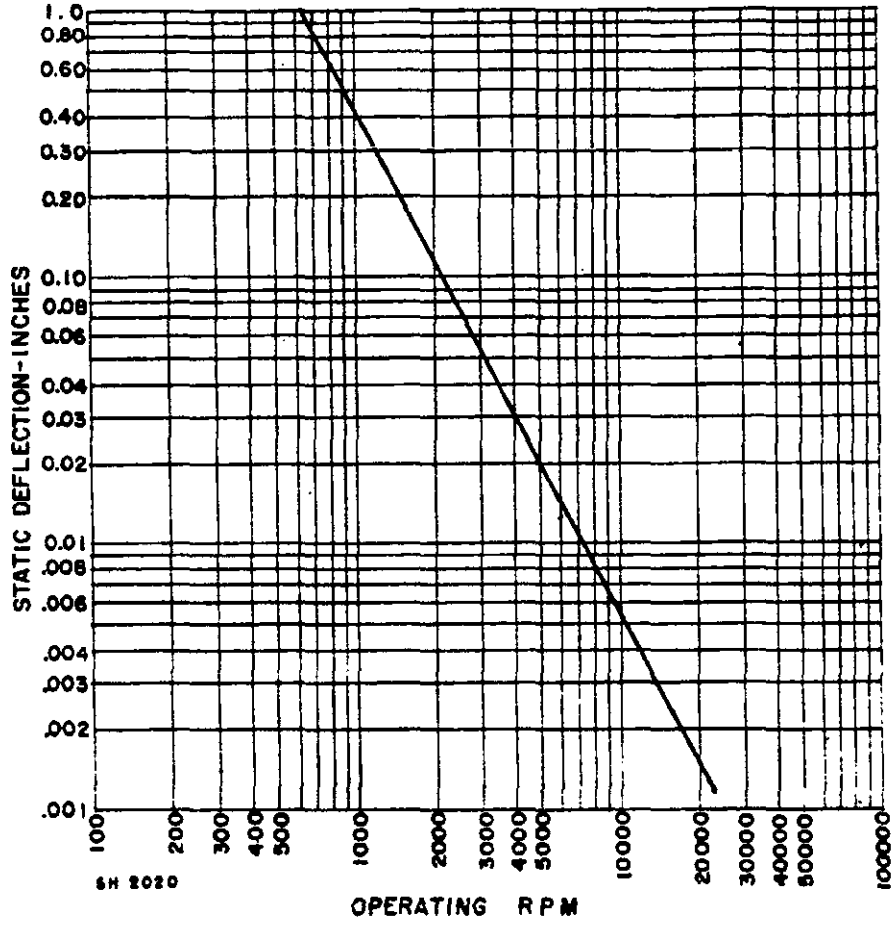


Figure 1 - Static deflection of elastic mounting for internally excited vibration test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

4.6.11 Generator no load saturation curve.- Sufficient data to plot a no load saturation curve for the generator shall be obtained. A copy of this curve shall be included in the test records. Points for this curve shall be taken as follows: Four below 60 percent of rated voltage; one between 60 and 90 percent; one at rated voltage; and three between rated voltage and full field excitation. Readings shall always be taken with increasing values of field current and when it is necessary to reduce the field, it must be reduced to zero and then increased to the desired value. Data recorded for each point shall consist of field voltages and currents, speed, generator voltage (each of the 3 phase voltages and average of the 3 phase voltages in the case of 3 phase generators), motor input voltage and motor input current. The motor generator shall be operating at rated speed and with rated voltage applied to the motor during this test.

4.6.12 Synchronous impedance curve.- Sufficient data to plot a synchronous impedance curve for the generator shall be obtained. A copy of this curve, which may be plotted on the same sheets as the saturation curve, shall also be included in the test records. Six readings from 150 percent rated current shall be recorded.

4.6.13 High impact shock test.- The test shall be as specified for type A in MIL-S-901 and as specified herein:

(a) The motor generator and control equipment shall be energized during the test; and may be tested simultaneously or separately. All equipment or parts shall be mounted on the shock machine in a manner simulating the actual installation on board ship. The weight designation of the shock test shall be as required by the combined weight of the equipment being tested. The average generator voltage and frequency shall be recorded before and after each blow. No adjustments or change in input or output shall be made between blows. Grade 2 bolts in accordance with MIL-B-857 shall be used to hold down the equipment during the shock test. When motor generators are tested at a 30 degree inclination on the medium weight shock test machine, the centerline of the shaft shall be parallel to the inclination.

(1) If tested separately, the following procedures are acceptable:

(a) The motor generator may be energized by a source other than its intended control equipment.

(b) The control equipment may be tested with a simulated load rather than controlling its intended motor generator. Data shall be recorded before and after each blow to demonstrate the equipments' ability to meet the regulation requirements of 3.2.9.

(b) Failure to perform principal function.- Any of the following failures shall be considered as cause for rejection of the equipment:

(1) Breaking of any parts.

(2) Appreciable distortion or dislocation of any parts such as mounting feet, poles, coils, brushes, brush holders, brush holder studs and bearings.

(3) A value of insulation resistance of the motor generator or any of the control equipment (corrected to 25°C.) less than that permitted by 3.2.5.

(4) Objectionable noise.- Any noticeable increase or decrease in noise of the motor generator when operating at rated speed should be investigated and its cause determined. However, if disassembly is required for this check, it should be made during the test specified in 4.6.7.

(5) Low dielectric strength.- The motor generator and control equipment shall be operated to obtain approximately rated temperature rise of iron and windings, then disconnected from its load and its source of power and a dielectric strength test made to check the condition of insulation. This dielectric strength test shall be made in accordance with MIL-E-917 except that it shall be made with an applied voltage equal to 65 percent of that specified.

(6) Failure to meet twice the internally excited vibration limit specified in 3.4.5.2 after completion of the shock test.

(7) A shift in regulated voltage or frequency of more than plus or minus 0.25 percent.

(8) Failure to meet the voltage and frequency regulation test specified in 4.6.16.2 after completion of the shock test.

(9) The equipment shall be operated at full load for 2 hours and its overall performance checked Any noticeable increase in heating, or other abnormal operation noted and investigated to determine its cause.

(10) Failure to pass inspection.- After all the checks and tests of 4.6.13 (b) (1) through (9) have been conducted, the motor generator and control equipment shall be disassembled and thoroughly inspected for damage. The extent of disassembly need be only to the point where the condition of the various components can be easily observed.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the currentversion before use.

(c) Disposal of shock-tested equipment. -
(1) Equipment which has been subjected to the high impact shock test and has failed to perform any of the principal functions specified herein will not be acceptable, either in whole or any of the parts.
(2) Equipment which has been subjected to the high impact shock test and has successfully performed all the principal functions specified herein will be acceptable under a contract or order providing that ball bearings are replaced in the motor generator.

(d) The place at which the shock test shall be conducted. - Shock tests may be performed at the contractor's plant, commercial laboratory, or Government laboratory which is suitably equipped to perform these tests. In either case, the tests are to be conducted at the expense of the contractor. Where shock tests are conducted at a Government laboratory, copies of all master drawings shall accompany the equipment.

(e) The number of generators to be tested. - Unless otherwise specified in the contract or order, one motor generator of the longest core length in each diameter and of similar construction shall be shock tested. If the manufacturer desires, he may submit a motor generator frame of less than the longest core length for the shock test. However, satisfactory compliance with the shock requirements will only determine conformance of motor generators having a core length equal to or less than the core length of the motor generator tested.

(f) Shock extension. -
(1) Motor generator shock test approval may be extended from the longest core length (same outside diameter and similar construction) to a motor generator of a shorter core length. The manufacturer shall submit a letter of request together with the drawings of each unit. Upon receipt of command or agency approval, the manufacturer shall note on the master plan the test number and date of original shock test, the inspector's approval letter, serial number and date and a statement of shock extension including command or agency approval letter.
(2) The extension of shock test from one unit of control equipment to another shall be based on comparison of size, weight, and component distribution. The manufacturer shall submit a letter of request together with the drawings of each unit. Drawing notation shall be as specified in 4. 6. 13 (f) (1).

4. 6. 14 Regulated voltage adjustment range and steps. - The tests for compliance with 3. 2. 12 shall be conducted as follows:

(a) With rated voltage applied to the motor and the generator operating at no load under automatic operation.
(1) Record the maximum and minimum voltage adjustment.
(2) Record the voltage for three consecutive steps for at least 5 points over the range specified.
(b) Repeat (1) and (2) above for rated load at rated power factor.
(c) The maximum and minimum limits of adjustability shall be recorded for the input variations specified in 3. 2. 1(a)(6).
(d) If remote sensing is specified (see 6. 1), this test shall be conducted under the conditions specified in 4. 6. 17(a).

4. 6. 15 Regulated frequency adjustment range and steps. - The tests for compliance with 3. 2. 16 shall be conducted as follows:

(a) With rated voltage and frequency applied to the motor and the motor generator operating at no load under automatic operation.
(1) Record the maximum and minimum frequency adjustment.
(2) Record the frequency for three consecutive steps for at least five points over the range specified.
(b) Repeat (1) and (2) above for rated load at rated power factor.
(c) The maximum and minimum limits of adjustability shall be recorded for the input variations specified in 3. 2. 1(a)(6).
(d) If remote sensing is specified (see 6. 1), this test shall be conducted under the conditions specified in 4. 6. 17 (a).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

4.6.16 Automatic voltage and frequency regulation. -

4.6.16.1 Periodic test. -

(a) The test shall be started with the rotating and control equipment at room temperature.
(b) Start the equipment with nominal voltage and frequency applied to the motor and with no load on the generator. Adjust the average voltage and the frequency to rated. No further adjustment shall be made to the voltage and frequency adjusting control or any other part of the regulators during the remainder of this test. The first set of readings shall be taken within 2 minutes after the equipment is energized. The generator voltage and current (or each of generator phase voltages, average of the phase voltages, and each of three output line currents in the case of three phase generators), kw output, speed, frequency, generator field voltage and current and motor input voltage frequency and current (each line) shall be the minimum data recorded for each set of readings taken during the tests.
(c) Apply full load to the generator and permit the equipment to reach its full load temperature rise before proceeding with the remaining parts of this test. A set of readings shall be taken every 30 minutes during warmup.
(d) A set of readings shall be recorded for each of the following tests specified in table XI.
(e) Repeat tests 1, 10, 13, 22, 25, 34, 37 and 46 of table XI for maximum input frequency.
(f) Repeat tests 3, 12, 15, 24, 27, 36, 39 and 48 of table XI for minimum input frequency.

Table XI - Automatic voltage and frequency regulation test conditions.

Percent of rated kw load	Power factor of load (lagging)	Motor input voltage and frequency (see 3.2.1(a)(6)	Test no.
0	-	Min. (minimum)	1
		Nom. (nominal)	2
		Max. (maximum)	3
25	0.8	Min.	4
		Nom.	5
		Max.	6
	1.0	Min.	7
		Nom.	8
		Max.	9
50	0.8	Min.	10
		Nom.	11
		Max.	12
	1.0	Min.	13
		Nom.	14
		Max.	15
75	0.8	Min.	16
		Nom.	17
		Max.	18
	1.0	Min.	19
		Nom.	20
		Max.	21
100	0.8	Min.	22
		Nom.	23
		Max.	24
	1.0	Min.	25
		Nom.	26
		Max.	27
75	0.8	Min.	28
		Nom.	29
		Max.	30
	1.0	Min.	31
		Nom.	32
		Max.	33

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

Table XI - Automatic voltage and frequency regulation test conditions (cont'd.).

Percent of rated kw load	Power factor of load (lagging)	Motor input voltage and frequency (see 3.2.1(a)(6))	Test no.
50	0.8	Min.	34
		Nom.	35
		Max.	36
	1.0	Min.	37
		Nom.	38
		Max.	39
25	0.8	Min.	40
		Nom.	41
		Max.	42
	1.0	Min.	43
		Nom.	44
		Max.	45
0	-	Min.	46
		Nom.	47
		Max.	48

Load	Generator terminal connection	Power factor of load (lagging)	Motor input voltage (see 3.2.1(a)(6))	Test no. 1/
Single phase 15 percent rated current	$T_1 - T_2$	0.8	Nominal	49
"	$T_1 - T_2$	1.0	Nominal	50
"	$T_2 - T_3$	0.8	Nominal	51
"	$T_2 - T_3$	1.0	Nominal	52
"	$T_3 - T_1$	0.8	Nominal	53
"	$T_3 - T_1$	1.0	Nominal	54

1/ Tests 49 through 54 are applicable to type I only. The generator shall be loaded with a 3 phase, 75 percent rated, 0.8 power factor load before the single phase, 15 percent rated current load is applied during tests 49 through 54.

The manufacturer may conduct any number of tests, in any sequence for a kw load. The tests must be conducted in order of ascending and descending load as shown in the table XI.

Tests 49 through 54 may be taken before or after tests 1 through 48, or interspersed with tests 16 through 21 and tests 28 through 33.

All voltages and frequencies recorded during the above tests shall be within the limits specified in 3.2.9 and 3.2.10 respectively.

4.6.16.2 Routine test. - This test shall be conducted in accordance with 4.6.16.1, except that only the following tests will be conducted in the order listed:

Test No.
3
22
47
17
23
41
51
15

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

4.6.17 Remote sensing - automatic voltage and frequency regulation. - When remote sensing is satisfied, the following regulation tests shall apply in lieu of those indicated in 4.6.16.

(a) The tests of 4.6.16.1 and 4.6.16.2 shall be performed with the load connected through an impedance such as to simulate a 7 percent rated voltage cable drop with voltage sensing at the remote load.

4.6.18 Voltage deviation and recovery time. - With supply voltage and frequency adjusted to rated values, an 0.8 p.f. load shall be applied and removed in steps as described in 3.2.10. Oscillograms shall be taken to determine compliance with 3.2.10 and 3.2.11.

If remote sensing is specified (see 6.1), this test shall be conducted under the conditions specified in 4.6.17(a).

4.6.19 Frequency deviation and recovery time. - With supply voltage and frequency adjusted to rated values, an 0.8 p.f. load shall be applied and removed in steps as described in 3.2.14. Oscillograms shall be taken to determine compliance with 3.2.14 and 3.2.15.

4.6.20 Parallel operation. -

(a) The parallel operation test shall be conducted with two motor generators as follows: With each motor generator connected to the same power supply, and the voltage regulator in automatic operation; adjust the control equipment so that rated voltage is obtained and the kw and kva loads of each generator are equal for a combined load of 25 percent. No further adjustment of the control equipment shall be made for the remainder of the test. The total load shall then be varied from 25 percent to 100 percent of combined kw and kva rating of the parallel generators at approximately rated power factor in three approximately equal steps and back to 25 percent load in the same manner. The actual difference in kw and kvar expressed in percentage of kva rating of one power supply shall be computed and recorded.
(b) The motor generators shall be parallel 180 degrees out of phase to demonstrate compliance with 3.2.23(b).

If remote sensing is specified (see 6.1), this test shall be conducted under the conditions specified in 4.6.17(a).

4.6.21 Short circuit characteristics. - The sustained short circuit current shall be measured by means of an ammeter and oscillograph to determine compliance with 3.2.25. When computing the I^2t value the steady state current measured with the ammeter shall be used, the time shall start when the current reaches its steady state value. When the circuit breakers are furnished by the contractor, the test shall be conducted with the circuit breaker locked in to prevent tripping during the test. There shall be no evidence of smoking, burning or burning odor as a result of this test. The voltage shall automatically return to within the regulation band limits when the short circuit is removed. Oscillograms of the fault current and voltage shall be taken.

If remote sensing is specified (see 6.1), this test shall be conducted under the conditions specified in 4.6.17(a).

4.6.22 Heating tests. -

4.6.22.1 General requirement. - All heating tests on the motor generator and control equipment shall be conducted at the same time. Simulated heating tests will not be permitted on the control equipment or motor generator. All heating tests shall be made on a production model. During heating tests of the control equipment, the enclosure doors shall be closed. In those cases where the loading of control circuit components may be greater at some low or intermediate load point than it is at rated load or overload, additional tests will be required to demonstrate that these components are not overheating at these lower loads.

If remote sensing is specified (see 6.1) this test shall be conducted under the conditions specified in 4.6.17(a).

4.6.22.2 Rated load heating test. - The equipment shall be operated at rated load at rated power factor and with nominal voltage applied to the motor until the temperature has stabilized in all components. However, in no case shall the duration of the test be less than 4 hours. During the first 3 hours, temperature measurements shall be made and recorded at 30 minute intervals. After that readings shall be made and recorded at 15 minute intervals. It shall be considered that stabilized temperatures have been reached,

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the current version before use.

when at least four consecutive readings not differing by more than 2 degrees are taken at 15 minute intervals. After shutdown, temperature rises of the rotating elements shall be determined in accordance with 4.6.22.5.3 and rise by resistance measurements made in accordance with 4.6.22.5.4. Maximum temperature rises recorded shall be within the limits specified in table I. For water cooled machines, the cooler shall be supplied with cooling water at rated flow with inlet water temperature at 85°F.

4.6.22.3 Special heating tests for type II equipment.- In addition to the tests specified in 4.6.22.2, the following tests shall be run for a period of not less than 4 hours:

(a) No load.
(b) Run No. 4 of 3.2.20.2.

The maximum rises recorded shall be within the limits specified in table II.

4.6.22.4 Data to be recorded.- The following data shall be the minimum recorded for each reading taken during the heating tests:

(a) Motor input voltage frequency and current.
(b) Output power (measured on load side of circuit breaker).
(c) Output frequency, phase voltages and line currents.
(d) Power factor.
(e) Room ambient.
(f) Air temperature at a minimum of four locations within the control equipment enclosures.
(g) Temperature of air entering and leaving all enclosures.
(h) Thermometer or thermocouple readings for all parts.
(i) Test data and calculations required for water cooler in accordance with MIL-C-19836 (water cooled machines only).

The temperature rise of all parts shall be determined by one of the methods specified in table II; however, the temperature rise of all motor and generator windings which have a resistance of 1.0 ohm or more shall be determined by the rise in resistance method. Where shutdown temperature of rotating elements (for windings less than 1 ohm) are taken in accordance with 4.6.22.5.3, extrapolation curves need not be included in the test report.

4.6.22.5 Measurement methods.- Measurement methods used in the heating tests shall be as specified herein. Detailed procedures of method I (thermometer) and method II (resistance) shall be as specified in MIL-E-917. In determining temperature rise, no correction shall be made for barometric pressure, humidity, or for any differences in heat transfer characteristics between the test ambient temperature and the specified 50°C. ambient temperature.

4.6.22.5.1 Ambient temperature.- The ambient temperature shall be measured by means of two or more thermometers placed at different points around each the motor generator and control equipment (total 4 minimum). The thermometers shall be placed at a distance of 3 to 6 feet from the equipment and at a height level with the mid point of the equipment. The ambient temperature thermometers shall be protected from drafts, and from heat radiation, from equipment under test or outside sources. They shall be inserted in heavy oil filled cups of not less than 1 inch in external diameter and 2 inches in height. The value to be adopted for the ambient temperature during the tests shall be the mean of the readings of the several thermometers, placed as above, taken at four equal intervals of time during the last quarter of the test. A motor generator power supply may be tested at any convenient ambient temperature above 10°C, but the conditions in the testing room shall be such that the ambient temperature will not vary greatly during test. A variation of 10°C, or more, during a period of 6 hours, or proportional change for runs of shorter durations shall in no case be exceeded. If the ambient temperature is very irregular during the run or changes rapidly at the end, the test shall be repeated.

4.6.22.5.2 Potted components.- The temperature rise of all encapsulated or potted components shall be determined by embedding thermocouples in the samples to be used for this test.

4.6.22.5.3 Rotating elements.- The temperature rise of collector rings, and generator rotating field or armature windings shall be determined as follows: On shutdown, preheated thermometers (preheated to a temperature approximately equivalent to the temperature rise permitted for the applicable part plus the room ambient at the time) shall be placed on the respective parts in accordance with method I procedures

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

of MIL-E-917. Every precaution shall be taken to reduce to a minimum the period of time elapsing between the stopping of the machine and the application of the thermometers. A curve should be plotted with temperature readings as ordinates and time as abscissa. The curve should be extrapolated back to the instant of shutdown. The temperature at the instant of shutdown shall be recorded and the temperature rise determined. This rise shall be in accordance with the limits specified in table I. In addition, the temperature rise of all rotating windings which have a resistance greater than 1.0 ohm shall be determined by method II of MIL-E-917.

4.6.22.5.4 Stationary elements of motor generator.- The temperature rise of motor and generator stationary coils, laminations or field poles and the outer races of both bearings shall be measured by method I of MIL-E-917. In addition, the temperature rise of all coils which have a resistance greater than 1.0 ohms shall be measured by method II of MIL-E-917 shutdown.

4.6.22.5.5 Qualified products.- Qualified products such as rheostats, resistors, circuit breakers, motor starters, capacitors, circuit breakers, instruments, and instrument transformers have previously been tested and are satisfactory when used within their rating and provided there is no excessive rise in air temperature within the complete control equipment enclosure. When all of the control equipment is integrated within a single enclosure, the rise of all transformer coils, motor controller coils, resistors, rheostats, case temperatures of capacitors, and air temperature within 1/2 inch of all instruments shall also be measured by method I of MIL-E-917 during the heat runs and be within the limits specified in table I.

4.6.22.5.6 Other components.- The temperature rise of components not covered by 4.6.22.5.2 through 4.6.22.5.5 shall be measured as follows:

(a) Transformer windings, magnetic amplifier windings, and so forth, method I of MIL-E-917.
(b) Semiconductor devices.- The ambient temperature of semiconductor devices shall be measured by method I of MIL-E-917 within 0.5 inch of the device on the low temperature side (side from which air starts passage over device). The case or cell temperature required for selenium rectifiers and other semiconductor devices relying upon a cooling fin or other surface to dissipate heat from the case to the ambient shall be measured by placing a thermocouple in direct contact with the case (on center cell near through bolt or stud of a selenium rectifier stack). Position on the case shall be on the high temperature side of the case. In no instance shall the thermocouple be placed on the cooling fin or other heat dissipating surface. The selenium cell or plate is an exception since the rectifying junction is distributed across the cell.

4.6.23 Insulation resistance (hot).- Immediately following the heat run, measurements of insulation resistance shall be taken. (For details of test see 4.6.5.)

4.6.24 Efficiency.- The efficiency of the motor generator power supply shall be measured for 1/4, 2/4, 3/4, and full load. Motor input voltage, frequency and current (each line), kw input, field voltage and current for generator fields, generator voltage (average), generator line currents measured at the generator terminals, output line current measured at the load side of the circuit breaker, load power factor kw output, generator frequency, and the calculated overall efficiency shall be the minimum data recorded for each load point. A detailed breakdown of losses will not be required. However, if an analysis is made for the contractor's own design files, it is requested that a copy of this breakdown of losses be included in the test report. The data taken during 4.6.16 may be used to compute the efficiencies.

4.6.25 Harmonic analysis.- A harmonic analysis shall be made with the equipment operating at no load, and full load unity power factor. All harmonics up to the 40th harmonic shall be recorded. Harmonics of less than 0.1 percent may be disregarded in the determination of the total operation when the harmonic analysis is made.

4.6.26 Amplitude modulation.- Amplitude modulation of the generator output voltage shall be measured to determine compliance with 3.2.18. This test shall be conducted with the generator voltage and frequency at rated value. The percentage of amplitude modulation shall be recorded for no load and full load unity power factor.

4.6.27 Deviation factor.- The deviation factor shall be determined with the equipment operating at no load, full load rated power factor, and full load unity power factor. The voltage and frequency regulator

Source: https://assist.dla.mil -- Downloaded: 20261O-19T06:25Z
Check the source to verify that this is the current version before use.

shall be in operation when the deviation factor is measured. The deviation factor shall be measured by either of the following methods:

(a) By oscillograph, or photograph of an oscilloscope trace, produce the generator waveform with at least 4 inches peak amplitude. Overlay the generator waveform curve with a pure sine wave so that the difference between them is minimum.

$$\text{Percent deviation factor} = \frac{\text{max. vertical difference (100)}}{\text{peak value of sine wave}}$$

(b) Photograph the lissajous figure of the waveform produced on an oscilloscope. Enlarge the photograph to be at least 8 inches minimum diameter and overlay it with a circle so that the difference between them is minimum.

$$\text{Percent deviation factor} = \frac{\text{max. radial difference (100)}}{\text{radius of circle}}$$

4.6.28 Line unbalance.-

(a) Type I.- The data recorded during tests 49 through 54 of 4.6.16 shall be used to determine compliance with the line unbalance requirements specified in 3.2.20.1. The maximum difference in line voltages for each of the load points shall be computed and recorded. The voltage values used shall be measured at the output terminals of the power supply (load side of generator circuit breaker) and not at the generator terminals. When remote sensing is furnished, the voltage values for tests 49 through 54 of 4.6.16 may be measured either at the generator output terminals or at the load.

(b) Type II.- With supply voltage and frequency adjusted to rated values, each of the unbalanced load conditions listed in 3.2.20.2 shall be applied for one hour. During each of these runs, the voltage on each line of the output and the temperature rise within the rotating and regulator units shall be recorded.

4.6.29 Voltage balance.- The data recorded during tests 1 through 48 of the regulation test 4.6.16 shall be used to determine compliance with the voltage balance requirements specified in 3.2.21. The maximum difference in line voltages for each of the load points shall be computed and recorded. The voltage values used shall be measured at the output terminals of the power supply (load side of the generator circuit breaker) and not the generator terminals.

4.6.30 Overvoltage limitation.- Overvoltage limitation features shall be tested as follows:

(a) A component failure shall be simulated (by adjusting the rheostat, opening the sensing leads, opening a control or bias lead, or some other means) to show that the voltage will be limited to a value within the overvoltage range. Voltmeter readings and oscillograms shall be taken of the output voltage to determine compliance with 3.5.4.8 (d). The motor generator shall be operating at nominal input voltage rated output voltage and frequency and 50 percent rated load before the test. There shall be no damage to the equipment as a result of these tests.

4.6.31 Starting tests.- At the instant of start, the locked rotor voltage, current and watts of the motor shall be recorded.

4.6.32 Component ratings.- The inspector shall ascertain that the rating of all components such as resistors, rectifiers, capacitors, transformers, instruments, rheostats, and so forth, is as shown on the approved master drawing.

4.6.33 Repair parts.- Repair parts shall be subjected to the following tests as applicable:

(a) Insulation resistance.
(b) Dielectric.
(c) Creepage and clearance.
(d) Materials.
(e) Resistance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

4.6.34 Curves and oscillograms.-

(a) Curves shall be drawn from the oscillograms showing voltage and frequency transient response and dip and rise as a function of time.
(b) All oscillograms shall be scaled and properly identified.

4.6.35 Electrical balance of motor.- Line currents measured during the rated load heating test may be used to indicate electrical balance of motor (see 3.2.26).

4.6.36 Effect on power source.- The manufacturer shall demonstrate compliance with the requirements of 3.5.4.9(e) by one of the following methods:

(a) Simulate a ship's electrical system which has the source impedance specified in 3.5.4.9(e). Measure the current modulation, and the voltage deviation factor at no load and full load. Power pulses shall be measured at no load and full load by a watt transducer or other method approved by the procuring activity.
(b) When the motor generator is operating from a source whose impedance is known, measure the current modulation, and the voltage deviation factor at no load and full load. By calculation determine the no load and full load values of the current modulation, and the voltage deviation factor when applied to a system with the source impedance specified in 3.5.4.9(e). Power pulses shall be measured at no load and full load by a watt transducer or other method approved by the procuring activity.

4.7 Inspection of preparation for delivery.- The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Motor generators.-

5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during shipment and storage or overseas shipment, which may be required when procurement is made:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. -

5.2.1.1.1.1 Motor generators. - The motor generators shall be preserved and packaged level A in accordance with MIL-P-16298. When the size and mounting arrangement of the equipment does not permit enclosure within a water-vaporproof barrier as required in a method II package, packaging shall be accomplished by the alternate method specified in MIL-P-16298.

5.2.1.1.2 Level C. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage. This level may conform to the supplier's commercial practice.

5.2.1.2 Packing. -

5.2.1.2.1 Levels A and B. - The motor generators shall be packed level A or B, as specified in accordance with MIL-P-16298.

5.2.1.3 Marking. - In addition to any special marking requirements, marking for shipment shall be in accordance with MIL-STD-129.

5.2.2 Repair parts and tools. - Repair parts and tools shall be preserved, packaged, packed and marked for the level of shipment in accordance with MIL-P-16298.

5.2.3 Manuals. - Manuals shall be packaged, packed and marked for the level of shipment in accordance with MIL-M-15071.

5.2.4 Drawings. - Drawings shall be packaged, packed and marked for the level of shipment in accordance with MIL-D-1000/2.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and arrangement of motor generator (see 1.2 and 3.5.2).
(c) Whether generator size required is 5, 10, 15, 30, 60, 100, 200 or 300 kw (see 3.2.1 and table IV). (If kw rating other than those listed is desired, rating size and maximum overall dimensions of control cabinets and MG unit plus mounting dimensions shall be given.)
(d) Whether remote sensing is required (see 3.2.8, 3.2.25, 3.5.4.8, 4.6.14, 4.6.15, 4.6.18, 4.6.20, 4.6.21 and 4.6.22.1).
(e) Whether water cooling is required (see 3.3.5, 3.3.5.2 and table IV, note 4).
(f) Maximum value of inrush current upon starting (see 3.5.4.9(d)).
(g) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy-SH
(Project 6125-N086SH)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

APPENDIX

INSPECTION CHECK LIST FOR IN-PROCESS INSPECTION

10. SCOPE

10.1 This appendix lists those items which should be included in the in-process inspection of motor-generator power supplies including control equipment furnished in accordance with this specification. This list is not an all inclusive list and additions and deletions as applicable can be made by the manufacturer subject to the approval of the Government.

20. APPLICABLE DOCUMENTS

20.1 This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 To facilitate the in-process inspection procedure the manufacturer shall prepare a manufacturing flow chart showing the normal flow of material, parts, subassemblies, and assemblies through the manufacturing process. The manufacturer shall indicate on the flow chart the points at which the manufacturer conducts in-process inspection and these points shall be designated as inspection stations.

30.1.1 For each inspection station the manufacturer shall prepare an inspection check sheet listing the items inspected at that station. Where appropriate, a standard sketch shall be included in the check sheet showing dimensions to be checked and the allowable tolerances, finish requirements, and so forth. Space shall be left on the sketch or on the check sheet to record the results of dimensional, visual or other inspection.

30.1.2 A column shall be included on the check sheet for the initials or stamp of the manufacturer's representative conducting the inspection. A space shall be left on the inspection check sheet for the Government inspector's initials or stamp.

30.1.3 The inspection check sheet shall be suitably protected against damage or staining and will be physically attached to the item being inspected. When the component is assembled, the inspection check sheet for the component shall be attached to the assembly or otherwise made available.

30.1.4 The words "approved drawing" as used herein mean a required drawing (type I drawing), which is approved by the cognizant approval activity, or a manufacturer's shop drawing if needed to depict details for in-process inspection not shown on a required drawing. Manufacturer's shop drawings which agree with approved required drawings may also be used for in-process inspection. Inspection requirements are not to be waived or reduced by selection of drawings.

30.2 General inspection items. - General inspection items shall be as follows:

(a) Material is as specified on approved drawing. Material was ordered in accordance with applicable material or component specification and was inspected in accordance with the requirements of the material or component specification.
(b) All welding is done by qualified welders and is in accordance with the approved drawing. There is no evidence of nonfusion, weld cracks, under-size welds, incomplete welds, heavy porosity, weld splatter or slag.
(c) All soldered connections are solidly bonded; there are no cold soldering, no rosin joints, no corrosive flux, no fractured joints, or excess solder. Satisfactory connections were made prior to soldering. Bolted connections include approved locking devices and are secured against vibration. Solderless connectors are properly crimped and the connector and crimping tool are proper size.
(d) Finished castings are as shown on approved drawing and are clean and free of molding sand, cracks, blow holes, splits and deformations. Sufficient material is allowed for machining. Casting defects have not been covered by unauthorized repairs.
(e) Machining is as shown on approved drawing. The surfaces, including mating surfaces, as applicable, are smooth, square and are free of burrs, sharp edges, tool marks, chatter marks,

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

and scratches and damage due to handling. Surface finish is as shown on approved drawing, and there are no tool marks except those normally associated with the indicated surface finish. There are no flaws exposed as the result of machining.

(f) No parts are Government surplus or have been previously used or reclaimed.

(g) All items including hardware (nuts, bolts, lockwashers, and so forth) are made of corrosion resistant material or are given a corrosion resistant treatment.

(h) The Government inspector may require a coil or winding to be cut apart to see the extent of varnish treatment and filling if there is any question as to the effectiveness of varnish treatment used.

(i) All bolts, nuts, set screws and other fasteners are secured in a manner which will preclude loosening in service. All locking devices shall meet the requirements of MIL-E-917.

(j) Insulation creepage and clearance distances are in accordance with approved drawing. Creepage distances are not achieved by means of cemented or butt joints.

30.3 Motor generators.-

30.3.1 Items to be inspected prior to assembly.- Items to be inspected prior to assembly shall be as follows:

(a) Shafts.-

(1) Items (a), (b), (e) and (f) of 30.2 apply.
(2) All dimensions, tolerances, keyways, fillets, shoulders, and surface finish are as shown on the approved drawing.
(3) Eccentricity tolerances including out of round for the bearing seats, bearing shoulder, collector ring seats, and the seats for the rotating armatures or rotors are as shown on the approved drawing.
(4) After welding on appurtenances, shaft has been stress relieved as shown on the approved drawing.
(5) The shaft was not built up in any way to correct errors or reclaim material (unless specific case basis approval of the command or agency concerned is obtained in advance).
(6) All shoulders are square as shown on the approved drawing.

(b) Collector rings.-

(1) Items (a), (b), (d), (e), (f), and (j) of 30.2 apply.
(2) All dimensions and dimensional tolerances of rings, sleeve and sleeve insulation are as on the approved drawing.
(3) Manufacturing and assembly processes used are exactly as specified on the approved drawing. Specific temperatures and curing cycles are as shown on the approved drawing.
(4) Brush contact surfaces are smooth, free of sharp edges, burrs, porosity, deep tool marks and chatter marks. Slip ring grooves have no burrs.
(5) Terminal studs are fixed to the rings in a positive manner as shown on the approved drawing.
(6) Terminal stud insulation is as shown on the approved drawing.

(c) Fans.-

(1) Items (a), (b), (d), (e), (f), (g), (h), (i), and (j) of 30.2 apply.
(2) Fans, except those covered by 30.3.1(d)(5) are balanced.
(3) All dimensions and the number of blades are as shown on the approved drawing.
(4) The fan surfaces which move the air are free from any irregularities, surplus weld material or any other projections which may be a source of airborne noise.
(5) Fans which break down into several parts are indexed in a manner such that they can be assembled in one way only.

(d) End brackets and bearing housings, end caps and frames.-

(1) Items (a), (b), (e) and (f) of 30.2 apply.
(2) Sufficient metal is available for drilling and tapping.

Source: https://assist.dla.mil -- Downloaded: 202640-19T06:25Z
Check the source to verify that this is the currentversion before use.

(3) All drilling, tapping and bolt centers are as shown on the approved drawing. Holes are clean, free of chips and are drilled straight. There are no burred threads. Holes are spot faced and are located so that edge distance is adequate in accordance with the approved drawing.
(4) All rust and grease is removed before painting. End brackets and frames fabricated and stored for periods prior to assembly were cleaned, and given a coat of primer or rust preventive before storing.
(5) The end brackets, except mating surfaces, such as bearing housings and rabbets, are cleaned, primed and painted on the inside surface prior to assembly. The inside of the frame is primed and painted if it did not receive at least two dips and bakes during the varnish treatment of the stationary electrical components.
(6) Mating surfaces of brackets and frame are concentric as shown on the approved drawing.
(7) All dimensions, dimensional tolerances and concentricities are as shown on the approved drawing.
(8) Bearing housings have sufficient metal to permit redrilling and bushing of the housing.

30.3.2 Items to be installed during assembly. -

(a) General. -

(1) Items specified in 30.2 apply.
(2) All parts are interchangeable and no selective matching of parts is necessary.
(3) Assembly and disassembly of all equipment can be done without the use of special tools except as otherwise specified in the contract. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

(b) Items to be checked during winding and assembly of rotating elements. -

(1) Keys as shown on the approved drawing are used to prevent rotational movement of all rotating parts, (such as collector rings, commutators, spiders, rotors or armatures).
(2) When rotating parts are pressed on the shaft, the pressure required to press on these parts is as shown on the approved drawing.
(3) When rotating parts are shrink fitted on the shaft, the interference shrink fits of the parts are as shown on the approved drawing.
(4) Axial movement of all parts is prevented by the methods shown on the approved drawing.
(5) Fans are secured to the shaft as shown on approved drawing.
(6) Coil connections for rotating armature, a.c. generators and all equalizer connections are insulated adequately as shown on the drawing.
(7) Bearings are of the size and type as shown on the approved drawing.
(8) The inner races of ball bearings are secured to the shaft by means of shaft shoulders, lock washers and nuts or by the opposed shoulder method as shown on the approved drawing.
(9) Lamination size and stacking are as shown on the approved drawing.
(10) Slot or ground insulation are as shown on the approved drawing.
(11) Slot or ground insulation extends beyond stacking as shown on the approved drawing.
(12) Wire size and type used are as shown on the approved drawing.
(13) Performed coils formed in accordance with the dimensions shown on the approved drawing and given at least one varnish treatment with clear baking varnish before installation in the slots.
(14) Undue force is not required to put slot wedge in place and insulation is not damaged or pushed out of place as the wedge is inserted.
(15) Slot wedges are of material and size as shown on the approved drawing. Length of wedge exceeds length of slot as shown on approved drawing.
(16) Coils are not loose after the wedges are in place.
(17) The wedge size is proper for the size and shape of the slot and there is no possibility that the wedge will cock in the slot and slip out.
(18) Coil connections are adequately insulated as shown on the drawing.
(19) Coil support and phase insulation is as shown on the approved drawing. Phase insulation is inserted between the phase coils and is shaped to fit coil configuration.
(20) Coil extensions are insulated and secured as shown on the approved drawing.
(21) Insulation materials used are as shown on the approved drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

(22) Dimensions of wound coils are as shown on the approved drawing.
(23) Rotating field poles are secured as shown on the approved drawing.
(24) If steel wedges are used to secure rotating field poles, the wedges are securely driven and positive assurance is provided that they will not work loose.
(25) Insulation is cleaned off and unless self fluxing wire is used coil wire ends are tinned before making soldered connections. Connections are properly soldered. There is no evidence of cold solder joints.
(26) Lead wires are of the type shown on the approved drawing.
(27) Windings are mechanically secured as shown on the approved drawing.
(28) When preformed coils are used, the coils are given one varnish treatment prior to insertion in the slots or on the poles and the completed winding assembly, including winding and pole is given the varnish treatments with approved clear baking varnish. In the case where preformed coils are not used the complete assembly is given at least three varnish treatments.
(29) Type of varnish, treatment and baking time cycle, and baking temperatures are as shown on the approved drawing.
(30) The treated windings and coils are clean, smooth and glossy with good bonding and filling. Varnish seals are complete and show no signs of cracks or breaks. The completed windings have no air bubbles, air pockets, voids or dry spots on the surface and are not soft or sticky.
(31) There is no sign of excessive varnish buildup on one side of the winding assembly and lack of varnish buildup on the other side. The thickness of the varnish on the winding assembly shall be uniform over the entire surface of the windings.
(32) Lead wires are insulated from ground and secured to prevent them from moving due to centrifugal force. Length and arrangement of wires shall permit ready repair; there is no aimless wiring resulting in "rats nests". Wiring agrees with the approved drawing.
(33) No glyptal or non-approved type of varnish or paint was put on any of the rotating elements.
(34) Coils or windings are not nicked or damaged during handling and processing.
(35) All completed rotating elements (including bearings), are dynamically balanced. The type of balance weights and method of securing are as shown on the approved drawing. They are secured in a manner such that they will not loosen in service. Balance weights are not attached in the airstream of the fan.
(36) Mica is under cut to specified depth. Copper segments are not damaged.

(c) Items to be checked during winding and assembly of stationary elements -

(1) Items 30. 3. 2(b) (10) thru (23) inclusive, (26) thru (32) inclusive, (34) and (35).
(2) Complete stator assembly or stator core is secured to frame or frame spider as shown on approved drawing.
(3) Force required to press complete stator assembly or stator core into the frame or frame spider is as shown on the approved drawing.
(4) Axial movement of the stator within the frame is precluded by means shown on the approved drawing.
(5) All lead wires are insulated from ground and secured within the frame with a suitable clamp or fastening device.
(6) All lead wires pass through the frame, enter the terminal box and are secured in such a manner to prevent chafing or abrasion as shown on the approved drawing.
(7) Terminal lugs of the type shown on the approved drawing are provided on the leads. Terminal lugs are properly crimped and there are not cut wire stands.
(8) Terminal boxes are of the type and size shown and are secured in the manner shown on the approved drawing.
(9) All leads are properly marked as shown on the approved drawing. It is possible to differentiate motor leads from generator leads.
(10) All nuts, bolts, screws and other hardware are tight and provided with locking devices as shown on the approved drawing.
(11) Insulation material as shown on the approved drawing is placed between field coil and frame prior to securing the pole.
(12) Field pole coil is not unduly stressed while tightening pole securing bolts. Insulation is not damaged in the tightening process.
(13) Each field coil received at least three separate dips and bakes with an approved clear baking varnish either before or after assembly into the stator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

(14) All salient field coil connections were either soldered or bolted and a solid joint was made.
(15) All salient field coil connections are adequately insulated as shown on the approved drawing.
(16) Terminal board material is as shown on the approved drawing.
(17) Terminal studs are secured to the board in a manner to preclude turning when attaching external cables.

30.3.3 Items to be inspected during final assembly. - Items to be inspected during final assembly shall be as follows:

(a) Items (a), (b), (c), (d), (e), (f), (g), (h), (i) and (j) of 30.2.
(b) End brackets properly match frames bearing housings and end caps. All holes align, there is no excess clearance, and no undue force is required to assemble parts.
(c) Coil ends do not protrude to a point where they may be damaged by assembly or removal of the end bracket and there is no possibility of interference with the rotating elements.
(d) Bearing outer races are secured in their housings as shown on approved drawing. The inside diameter of the retaining shoulders is within the dimensional tolerances shown on the approved drawing.
(e) There is no evidence of grease leakage past the close clearance non-rubbing seals into the motor generator or along the shaft.
(f) Spring washers, if used to provide preloading of the bearing, shall be selected and secured in the housing as shown on approved drawing.
(g) Seals of prelubricated bearings are not damaged and there is no evidence of grease leaking out of the bearing.
(h) Brush holder studs are secured as shown on the approved drawing.
(i) Brush rigging insulation is as shown on the approved drawing.
(j) Brush holders and springs are of the type and size shown on the drawing.
(k) Brush holders are secured as shown on the approved drawing.
(l) Brushes are of the manufacturer's grade designation shown on the approved drawing.
(m) Brushes have been properly seated on slip rings over a minimum of 90 percent of contact area of brush and excess carbon has been removed from the machine.
(n) Brush shunts are attached to the brush material by imbedding or riveting.
(o) Brush size is as shown on the approved drawing and fits properly in the holder without sanding or filing and brushes move freely in the brush holders in the direction parallel to the length of the brush.
(p) Brush tension is adjustable and is set as shown on the approved drawing. Spring tension can be readily measured by spring scale.
(q) Brushes are properly aligned parallel to the shaft and do not extend beyond the edge of the slip rings.
(r) Brush lug terminals are properly secured by screws which are not used for securing the brush holders to the rigging.
(s) Brush leads are furnished with lugs and are adequately secured.
(t) Air gaps have been adjusted to design value and are uniform within the limits shown on the approved drawing.
(u) Bearings have been lubricated with grease as specified on the approved drawing.
(v) Air baffles are securely attached and do not interfere with rotating elements.
(w) Lifting means is provided as shown on the approved drawing.
(x) Equipment enclosure is as shown on the approved drawing.
(y) Entire exterior, other than identification plates or shaft extensions is painted in accordance with MIL-E-917.
(z) Identification and instruction plates are as shown on the approved drawing; contain all required information and are legible.
(aa) Mounting feet are flat, square, and are as shown on approved drawing.
(bb) The overall dimensions, mounting dimensions, and location of terminal boxes are as shown on the approved drawing.
(cc) Where high tensile bolts or screws such as socket head types, are used, provision is made to prevent replacement with a lower tensile strength bolt or screw. There are no slotted head bolts or screws used to secure end brackets to the frame. Thread-cutting screws (self tapping) are not used to secure any part of the motor generator.
(dd) Spot facing is as specified in MIL-E-917.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

(ee) The motor generator, particularly the bearing housing, are clean and free of dirt, metal chips or other foreign materials.
(ff) The weight of the complete unit is as shown on the drawing.
(gg) Stationary lead wires do not contact rotating parts.

30.4 Control equipment. -

30.4.1 Items to be inspected on major parts. -

(1) The following items were ordered in accordance with the applicable specification as modified by the contract requirements; are of the size, type and rating shown on the drawing; and are qualified products:

(a) Resistors and rheostats.
(b) Instruments, instrument transformers, shunts and other associated accessories.
(c) Motor starters and associated push buttons.
(d) Circuit breakers.
(e) Capacitors.

(2) The following items are not necessarily qualified products but are of the size, type and rating shown on the drawing:

(a) Switches.
(b) Rectifiers.
(c) Molded terminal boards.
(d) All laminated materials such as glass mica glass, glass silicon glass, used for terminal boards, sub panels, and so forth.

(3) All materials used in the construction of reactors, magnetic amplifiers, and transformers other than instrument transformers are as shown on the approved drawing and comply with the material requirements as shown on the approved drawing.
(4) All windings of reactors, magnetic amplifiers, and transformers other than instrument transformers were given a minimum of three varnish treatments or one vacuum impregnation and bake followed by one dip and bake with an approved clear baking varnish in accordance with the procedures shown on the approved drawing.
(5) No black varnish, glyptal, paint, or other treatment was given to the insulation of reactors, magnetic amplifiers, and transformers other than instrument transformers after the last dip and bake with clear baking varnish.
(6) The resistance of all windings of all reactors, magnetic amplifiers, and transformers other than instrument transformers is as shown on the approved drawing. Ratio of turns is as specified on the approved drawing.
(7) The equipment enclosures are as shown on the approved drawing.

30.4.2 Items to be inspected on completed unit. -

(1) All components are mounted as shown on the approved drawing.
(2) All parts which require servicing, repair, replacement, or periodic adjustment during the life of the equipment are readily accessible.
(3) All components are marked with identifying symbols and these symbols agree with those used on the approved drawing.
(4) All terminals of all components and terminal boards are marked with identifying letters and numbers or both and this identification agrees with those used on the approved drawing.
(5) Wire used for interconnecting components is of the type and size shown on the approved drawing.
(6) All wiring is neatly formed into groups and laced with noninflammable glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.
(7) Cable clamps used are of noninflammable material.
(8) There are no splices in any of the wires.
(9) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in either the open or closed position.
(10) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections and so that at least two replacements can be made if lugs are clipped off at the wire end.

Source: https://assist.dla.mil -- Downloaded: 202610-19T06:25Z
Check the source to verify that this is the currentversion before use.

(11) Cable entrance provisions are as shown on the approved drawing.
(12) Sufficient space is provided to bring external cables through these cable entrances, support them within the enclosure, and make connections at the appropriate terminals.
(13) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(14) Wires are not bent around sharp corners which may injure the insulation.
(15) All wires are connected by either bolted or soldered connections.
(16) All bolted connections are provided with locking devices.
(17) The soldered connections do not rely on the solder for mechanical strength. The wire was mechanically secured before soldering and the solder was used only to get a good electrical path.
(18) No more than 3 wires are soldered to any one terminal.
(19) Both ends of all wires are marked with designations as shown on the approved drawing.
(20) Hinged doors and panels do not bind when opening.
(21) A door stop and positioning device is furnished and functions satisfactorily.
(22) All identification and instruction plates are furnished as shown on the approved drawing.
(23) All information on identification and instruction plates is legible.
(24) A copy of the schematic and wiring diagram for the complete system as well as a copy of the operating instructions has been laminated in plastic and is stowed in a convenient location within the control panel.
(25) The location of the plastic laminated drawings and instructions is such that it does not interfere with convection air flow which may increase heating of components. In those cases where the location is questionable, heat runs of the control panels are run with the diagrams in place.
(26) The overall and mounting dimensions of the equipment are as shown on the approved drawing.
(27) The overall weight of the equipment is as shown on the approved drawing.
(28) The exterior and interior of the enclosure have been painted as shown on the approved drawing.
(29) All controls operate smoothly throughout their range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

INDEX

	Paragraph	Page
Access holes	3.4.13	14
Airborne noise	3.4.4, 3.5.2.4, 4.6.7	11, 17, 34
Ambient temperature	3.2.3, 4.6.22.5.1	3, 41
Ammeter	3.5.4.13	21
Amplitude modulation	3.2.18, 4.6.26	5, 42
Balance	3.4.5.1, 3.4.5.2, 4.6.10	11, 11, 35
Bearings	3.4.6	11
Bearing life	3.3.4(a)	7
Brush holders	3.4.11	13
Brush life	3.3.4(b)	7
Brush rigging	3.4.10	13
Circuit breakers	3.5.4.11	21
Coil connections	3.4.17	15
Collector rings	3.4.9	13
Component requirements and ratings	3.5.4, 4.6.32	19, 43
Control equipment	3.5	16
Cores	3.4.8	12
Creepage and clearance	4.6.1	33
Curves and oscillograms	4.6.34	44
Definitions	3.1	2
Deviation factor	3.1.8, 3.2.19, 4.6.27	3, 5, 42
Dielectric strength	3.2.24	6
Dielectric tests	4.6.6	34
Dimensions and weight	3.4.3, 3.5.3.2	10, 17
Drawings	3.7	23
Duty	3.2.2	3
Effect on power source	3.5.4.9(e), 4.6.36	20, 44
Efficiency	4.6.24	42
Electrical balance of motor	3.2.26, 4.6.35	6, 44
Electromagnetic interference	3.4.20	15
Enclosure	3.3.5, 3.5.3, 4.6.2	7, 17, 33
End brackets	3.4.12	14
End play	4.6.8	34
End stops	3.5.4.6	19
Excitation requirements	3.4.18	15
Excitation power	3.2.8	4
Fans	3.4.7	12
Feet	3.4.14	14
Field coils	3.4.16	15
Field poles and shoes	3.4.15	14
Frequency adjustment	3.2.16, 4.6.15	4, 37
Frequency deviation	3.2.15, 4.6.19	4, 40
Frequency (generator)	3.2.1(b)(4)	3
Frequency meter	3.5.4.14	21
Frequency (motor)	3.2.1(a)(4)	3
Frequency recovery time	3.2.14, 4.6.19	4, 40
Frequency regulation	3.2.13, 4.6.16	4, 38
Generator no load saturation curve	4.6.11	36
Grounding	3.2.6	3
Grab rods	3.5.3.8	18
Harmonic content	3.2.17, 4.6.25	4, 42
Heating	3.3.7, 3.5.3.13, 4.6.22	7, 18, 40
Identification and information plates	3.3.10, 3.5.6, 3.4.26	9, 22, 16
Inclined operation	3.3.11, 4.6.9	9, 34
Inrush current	3.5.4.9(d)	20
Inspection of preparation for delivery	4.7	44

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

INDEX (Cont'd.)

	Paragraph	Page
Instrument transformers	3.5.4.16	21
Insulation class	3.3.6	7
Insulation resistance	3.2.5, 4.6.5, 4.6.23	3, 34, 42
Interchangeability	3.3.3	7
Internally excited vibration test	4.6.10	35
Kilowatts	3.2.1(b)(1)	3
Lead identification	3.4.25	16
Life	3.3.4	7
Lifting provisions	3.3.9	9
Line unbalance	3.2.20, 4.6.28	5, 43
Manuals	3.8	30
Marking, permanent	3.4.27	16
Material restrictions	3.3.8	8
Mode of operation	3.2.7	4
Motor controller	3.5.4.7	19
Motor generator	3.4	9
Mounting	3.4.2, 3.4.6.5, 3.4.8.1, 3.5.3.1	10, 12, 12, 17
Operating and maintenance instructions	3.5.3.9	18
Ordering data	6.1	45/46
Overvoltage limitation	3.5.4.8(d), 4.6.30	20, 43
Parallel operation	3.2.23, 4.6.20	6, 40
Performance requirements	3.2	3
Periodic tests	4.5.1, 4.6.16.1	33, 38
Phase (generator)	3.2.1(b)(3)	3
Phase (motor)	3.2.1(a)(2)	3
Preparation for delivery	5	44
Quality assurance provisions	4	31
Relays	3.5.4.10	21
Reliability	3.3.2	6
Remote sensing	3.2.25, 3.5.4.8(e), 4.6.17	6, 20, 40
Repair parts	3.6, 4.6.33	23, 43
Resistance (cold and hot)	4.6.4	34
Routine tests	4.5.2, 4.6.16.2	33, 39
Securing terminal leads	3.4.24	16
Shaft	3.4.1	9
Shock	3.2.4, 4.6.13	3, 36
Shock extension	4.6.13(f)	37
Short circuit characteristics	3.2.22, 4.6.21	5, 40
Speed	3.2.1(a)(3)	3
Starting tests	4.6.31	43
Stray field reduction	3.4.19	15
Synchronous impedance curve	4.6.12	36
Synchroscope	3.5.4.15	15
Terminal boards	3.4.22	15
Terminal boxes	3.4.21	15
Terminal leads	3.4.23	15
Test reports and approval	4.4	33
Ventilation	3.5.3.10	18
Voltage adjustment	3.2.12, 4.6.14	4
Voltage balance	3.2.21, 4.6.29	5, 43
Voltage deviation (dip or rise)	3.2.11, 4.6.18	4, 40
Voltage recovery time	3.2.10, 4.6.18	4, 40
Voltage regulation	3.2.9, 4.6.16	4, 38
Voltage regulator	3.5.4.8	19
Voltmeter	3.5.4.12	21
Water cooling	3.3.5.2	7
Wattmeter	3.5.4.17	21
Weight	3.4.3, 4.6.3	10, 34
Wire end connectors	3.5.5.7	22
Wiring and connections	3.5.5	21

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION (YYMMDD)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T06:25Z
Check the source to verify that this is the current version before use.

DD FORM 1426